xMixing Control — Master User Manual

คู่มือมาตรฐานการใช้งานและเอกสารทางเทคนิค (User & Operator Manual + Technical Reference)

ระบบควบคุมกระบวนการผสม xMixing Control — Mitr Phol

รหัสเอกสาร	MAN-PRD-MIX-MASTER-001
Revision	01 — Master Edition
วันที่	25 มิถุนายน 2026
ผู้จัดทำ	ฝ่ายวิศวกรรม, IT, ฝ่ายผลิต

สารบัญ (Table of Contents)

บทที่ 1: คู่มือปฏิบัติงาน Operator

Work Instruction & Data Flow (xMixing Control)

เอกสารระบุลอจิกการทำงานและคู่มือปฏิบัติงาน (Work Instruction - WI) สำหรับ Operator ตลอดจนการทำงานร่วมกันระหว่าง **Scanner ↔ xMixing App ↔ PLC**

1. Data Flow Architecture (การรับส่งข้อมูล)

ระบบจะถูกขับเคลื่อนด้วย **“คน” เป็นผู้คุมจังหวะ** และ **“PLC” เป็นผู้ลงมือทำ**

```
sequenceDiagram
    participant SC as 🖮 Barcode Scanner
    participant APP as 🖳 xMixing App
    participant DB as    Database
    participant PLC as 🎛 PLC (S7-1200)

    Note over SC,APP: 1. Preparation & Validate
    SC->>APP: Scan Batch QR/Barcode
    APP->>DB: ตรวจสอบแผนผลิต & ดึงสูตร (Recipe)
    DB-->>APP: ส่งคืนสูตรทีละ Step ไปโชว์หน้าจอ HMI

    Note over APP,PLC: 2. Execution (Step-by-Step Handshake)
    APP->>PLC: 🚜 ยิง Send Step 1 (อุณหภูมิ, น้ำหนัก, เวลา)
    APP->>PLC: 🔘 ส่งคำสั่ง COMMAND: START
```

```
loop During execution
    PLC-->>APP:    Telemetry (RPM, Temp, Weight ปัจจุบัน) ทุกๆ 1s
end

PLC->>APP:    แจ้งจบการทำงาน STATUS: STEP 1 COMPLETE
APP->>DB:    Auto-save ประวัติการทำงาน Step 1 (รันนานแค่ไหน, ค่าจริงเท่าไหร่)

Note over APP,PLC: ระบบจะวิ่งไป Step 2 จนจบ Phase (หรือรอ Operator กดเริ่ม Phase ใหม่)

Note over APP,SC: 3. Manual QC & Validation (บาง Step)
APP->>SC: ⚠ บังคับ Scan ยืนยันวัตถุดิบ (ถ้าจำเป็น)
APP->>APP: ⚠ บังคับกรอก Brix / pH (ในขั้นตอน x1030/x1040)
APP->>DB:    Save Production & QC Record
```

---

2. Work Instruction (WI) ขั้นตอนการปฏิบัติงานของ Operator

🗔 ขั้นตอนที่ 1: การรับแผนและตั้งค่าเริ่มต้น (Preparation)

1. Operator เข้าสู่ระบบ (Login) ที่หน้าจอ HMI บนเครื่องจักร **Mixing Plant**
2. ไปที่เมนู **“Check for Production (x60)”**
3. เตรียมใบสั่งผลิต (Job Order)
4. ยก **Scanner** ขึ้นแสแกน QR Code/Barcode ที่อยู่บนใบสั่งผลิต
    - *ถ้าระบบหา Batch เจอ -> จะแสดงรายละเอียดสูตร (SKU), น้ำหนักที่ต้องการผลิตทางฝั่งขวา*
    - *ช่องสีเทา/สีฟ้าจะแยก Phase ให้อ่านง่าย*
5. ตรวจสอบชื่อสูตรและปริมาณให้ตรงกับใบจริง
6. กดปุ่ม **[GO TO PRODUCTION CONTROL]** ด้านขวาล่างเพื่อเข้าสู่หน้าคุมเครื่อง (x61)

⚙ ขั้นตอนที่ 2: เริ่มต้นรันเครื่องจักร (Start Batch)

1. ในหน้า **Mixing Control (x61)** หน้าจอจะแสดงหน้าปัดควบคุม (Gauges) เป็นค่า 0
2. ตรวจสอบสถานะว่าป้ายข้างๆ โชว์คำว่า 🟢 **PLC CONNECTED**
3. แจ้งพนักงานห้องเตรียมวัตถุดิบ หรือตรวจเช็คส่วนหน้าให้พร้อม (ระบบพร้อมโหลดน้ำ / น้ำตาลทราย)
4. กดปุ่ม ▶ **[START]** สีเขียวตรง Command Center
5. แอปฯ จะโยน **Step ที่ 1** ลงไปหา PLC และ PLC จะเริ่มทำงานทันที (ดึงน้ำ, ดันอุณหภูมิ)
6. Operator เฝ้าดูหน้าจอบอกสถานะ **Actual vs Setpoint**
    - *ไม่ต้องกด Start ทุกๆ Step ภายใน Phase* — PLC กับ App จะคุยกันเองว่าส่งของถูกไหม

⏸ ขั้นตอนที่ 3: กรณีฉุกเฉินหรือหยุดกลางคัน (Pause / Abort)

- หากต้องการดูอาการ หรือตรวจเช็คคุณภาพถัง แตะ ⏸ **[PAUSE]**
    - *PLC จะพักการปั่น และหยุดนับเวลา แต่จะไม่เคลียร์ค่า*
    - หากพร้อมทำต่อ กด ⏸ อีกครั้ง เพื่อทำต่อจากจุดเดิม
- หากสูตรผิด หรือเครื่องพัง ให้กด ⏹ **[ABORT]** สีแดง
    - *ระบบจะหยุดจ่ายไฟ และทำการล้างค่าในระบบ Operator ต้องไปยกเลิก Batch และออกหมายเหตุ*

ขั้นตอนที่ 4: การเติมส่วนผสมย่อย (Manual Dosing) & ใส่ค่า QC

1. ใน **Phase การละลายสาร (D1010/D1030)** ที่หน้าจอจะโชว์ว่าให้ใส่ส่วนผสมอะไร (เช่น Potassium Sorbate 0.5 kg)

2. Operator หยิบ **Scanner** มายิงบาร์โค้ดที่ถุงส่วนผสม *เพื่อยืนยันว่าหยิบไม่ผิดตัว* (Option เสริมในอนาคต)
3. เมื่อ PLC ปล่อยอุณหภูมิถึงจุด **Holding** หรือ **Cooldown (x1030, x1040)**
4. เครื่องจะหยุดรอการยืนยัน
5. Operator นำสายวัด หรือดึง Sampling ออกมาวัด **Brix** และ **pH**
6. กรอกค่าที่วัดได้จริง ลงในช่องสี่เหลี่ยมบน HMI
7. กด **[CONFIRM / NEXT_STEP]** วิ่งสเตปต่อไป

ขั้นตอนที่ 5: สิ้นสุดกระบวนการ (End Batch)

1. เมื่อครบ 35 Steps หรือจนกว่าครบทุก Phase ระบบจะขึ้นแจ้งพุชป๊อปอัป “ **BATCH COMPLETE”**
2. ระบบจะบันทึก Log ลง Database พร้อมทำรายงาน
3. Operator กดเปิดหน้าปัดล้างถังกวน หรือปั๊มออกไปยังถัง Holding ถัดไป
4. สิ้นสุดงาน (สามารถกดปริ้นท์ Log ท้ายกะได้จากปุ่ม 🖶 ปริ้นท์ด้านบนขวา)

---

3. สรุปความรับผิดชอบ (Who does what?)

ผู้รับผิดชอบ	หน้าที่
Operator	สแกน Job Order, กด START, เฝ้าระวังจอ (Monitor), ใส่ส่วนผสมมือ (Manual Dosing), กรอกค่า QC, ตัดสินใจกด STOP ฉุกเฉิน
💻 **xMixing App**	โหลดสูตร, แยกเป็น Step, ยิง Step ไป PLC, รับแจ้งจบ Step แล้วอัพโหลดใส่ Database (รันอัตโนมัติ 80%), ตรวจสอบ Tolerance อาราม
🎛️ **PLC**	อ่านค่าน้ำหนักเครื่องชั่ง, หมุนปั๊ม, จ่ายสตรีมให้ร้อน, คุม High Shear ตาม Step ที่รับมา
⌨️ **Scanner**	ลด Human Error ด้วยการสแกนเลขโค้ด / สแกนหลอดส่วนผสมก่อนโยนลงถัง

บทที่ 2: ขั้นตอนการสแกนวัตถุดิบ

Barcode Scanning & Verification Workflow

This diagram illustrates the step-by-step logic the application follows when an operator scans a QR code using the handheld scanner. It highlights the error-handling logic for various scanning mistakes, including the Wrong Box handling.

```
flowchart TD
    A([Scanner reads QR Code]) --> B{Is a Batch already loaded?}
    B -- No --> C[Load the scanned Batch]
    B -- Yes --> D{Is scanned code a completely new Batch ID?}

    D -- Yes --> E[Switch and Load the new Batch]
    D -- No --> F[Verify as an Ingredient Bag]
```

```
F --> G[Check local data for matching ingredient]
G -- Match Found --> H([Mark Ingredient as    Verified])
G -- No Local Match --> I[Send scan to Backend API]

I --> J{Backend Response}
J -- Success --> K([Mark Ingredient as    Verified])

J -- Failure --> L{What did the operator actually scan?}

L -- "The same Batch ID again" --> M([⚠ Yellow Warning: <br>'You scanned the Batch Label!'])
L -- "Unrecognized Bag for this Batch" --> N([✕ Red Toast Error: <br>'Ingredient not found'])
L -- "Bag from a completely different box" --> O([   Big Red Modal: <br>WRONG BOX!])

O --> P{Operator Decision}
P -- "I want to keep working on my current batch" --> Q([Close Modal & Continue])
P -- "I want to scan a different batch instead" --> R([Start New Batch / Reset])
```

บทที่ 3: หน้าจอ Mixing Control

🎛️ x61 - Mixing Control Page Concept (Plant 1)

หน้านี้คือ **หัวใจหลักของการดำเนินงาน (Operation Control)** เป็นหน้าจอที่ Operator ใช้คุมเครื่องจักรและตรวจสอบสถานะทุกอย่างระหว่างการผลิต โดยในเฟสนี้เราจะโฟกัสที่ **Plant 1 (Mixing Tank 1)** เป็นหลัก

---

1. หัวใจหลักของหน้า x61 (Core Concepts)

1. **Safety & Clarity FIRST** : ต้องเห็นค่าสำคัญชัดเจน (อุณหภูมิ, น้ำหนัก) และมีปุ่มฉุกเฉิน (ABORT) ที่เรียกใช้งานได้ทันที
2. **Step-by-Step Handshake** : หน้าเว็บจะเป็นคนจ่ายงาน (Recipe) ทีละบรรทัดให้ PLC และรอจนกว่า PLC จะทำสเตปนั้นเสร็จ แล้วจึงไปต่อ
3. **Actual vs Setpoint Comparison** : โชว์เปรียบเทียบค่าที่ควรจะเป็น (SP) กับค่าที่วัดได้จริงจากเซนเซอร์ตลอดเวลา
4. **Mandatory QC Stop** : ถ้าเป็น Step ที่เกี่ยวกับการเย็นลง (Cooldown/Final) ระบบต้องหยุดให้ Operator กรอกค่า Brix / pH ไม่งั้นไม่ให้จบ Batch

---

2. โครงสร้างหน้าจอ (UI Layout Design)

หน้าจอออกแบบมาให้ Operator เข้าใจง่าย แบ่งเป็นโซนชัดเจน (3 โซนหลัก):

▤ Zone A: Header & Status (ด้านบนสุด)

- วัตถุประสงค์: ข้อมูลแผนการผลิต
- ข้อมูลที่แสดง:
  - ชื่อ SKU และรหัสินค้า (เช่น S77S743200 - Cafe Amazon)
  - หมายเลข Batch และ Target Weight (เช่น 1200kg)
  - เชื่อมต่อ PLC สำเร็จหรือไม่ (แสดงป้าย ▩ PLC CONNECTED)

▧ Zone B: Live Telemetry Dashboard (ตรงกลางจอ)

- วัตถุประสงค์: แสดงสถานะเครื่องจักร Real-time ทุก 1 วินาที
- ข้อมูลที่แสดง (แบ่งเป็น 4 คอลัมน์):
    1. **Hopper Scale**: น้ำหนักของสารระเหย หรือสารปรุงแต่งที่รอโหลด
    2. **Mixing Tank (ถังหลัก)**: อุณหภูมิน้ำ, น้ำหนักรวม, ความเร็วใบกวน Agitator
    3. ⚡ **High Shear**: ความเร็วใบมีดบดละเอียด และอุณหภูมิมอเตอร์
    4. **Circulation**: ความเร็ว Flow Pump
- **จุดเด่น**: ในส่วนนี้จะมีป้ายตัวเลขเล็กๆ โชว์คำว่า `SP: 40°C` แปะไว้ข้างๆ `Actual: 42°C` ถ้าอยู่ในเกณฑ์ ±5°C จะเป็นสีเขียว ถ้าเลยเกณฑ์จะเป็นสีส้ม

▩ Zone C: Operation & Command Center (ด้านล่าง)

- วัตถุประสงค์: การควบคุม และบอกพิกัดว่าถึงขั้นไหนแล้ว
- ข้อมูลที่แสดง:
    - **Command Buttons**:
        - ▶ `START`: จ่ายงาน Step ปัจจุบันให้ PLC รัน
        - ⏸ `PAUSE`: สั่งให้ PLC ระงับการขยับ ทุกอย่างหยุดหมุน
        - ⏹ `ABORT`: เบรกฉุกเฉิน เคลียร์ระบบ
        - ⏭ `Force Next Step`: ข้ามสเตปแบบบังคับ (ใช้กรณีฉุกเฉิน/Manual Override)
    - **SKU Step List (Accordion)**: แกะกล่องทีละ Phase (เช่น Phase A1010) แล้วแสดงว่า PLC ตอนนี้บรรทัดไหนติดแถบสีสว่างๆ (Highlight) ให้รู้ว่า “กำลังกวนตัวนี้อยู่นะ”

---

3. ⚙ ลำดับการควบคุมจากหน้า x61 (Operation Flow)

1. **Initiate (โหลดข้อมูล)**:
    - เมื่อข้ามมาจากหน้า x60 ระบบจะดึง “ทุก Step” ใส่ตารางเก็บไว้
    - เชื่อมต่อ MQTT รอรับค่าจาก Plant 1
2. **Execute (การรันแบบปกติ)**:
    - Operator กด ▶ `START`
    - หน้า x61 ส่งข้อมูล *Step ที่ 1 เท่านั้น* ไปที่: `mixing/plant/1/step_cmd`
    - รอรับข้อความ `status: STEP_COMPLETE` จาก PLC (วนลูปแบบนี้จนจบ)
3. **The QC Trap (ขั้นตอนบังคับ)**:
    - เมื่อ Step ไหลไปจนถึงกลุ่ม Cooldown (`x1030`, `x1040`)
    - หน้าจอจะค้าง ไม่ Auto-advance
    - มีกล่อง Input โผล่มาให้กรอก: `Actual Brix [ ]` และ `Actual pH [ ]`
    - ต้องกรอกแล้วกด Save -> ระบบบันทึกลง DB -> ค่อยวิ่งต่อไป
4. **Conclusion (จบงาน)**:
    - เมื่อถึงบรรทัดสุดท้าย โชว์ข้อความสำเร็จ พิมพ์เอกสารสรุป Batch Report อัตโนมัติ

---

4. ลอจิกการคุยกับระบบหลังบ้าน (Backend API / Database)

เนื่องจากต้องบันทึกประวัติ (History) หน้า x61 นี้จะเป็นจุดเดียวที่ยิง API ลงฐานข้อมูล:

- **เมื่อทำ Finished 1 Step**:
    - `POST /api/records/step`

    - Payload: `{ batch_id, step_id, actual_temp_end, duration_seconds }`
- **เมื่อกด Confirm QC Check**:
    - `POST /api/records/qc`
    - Payload: `{ batch_id, phase_id, actual_brix, actual_ph }`
- **เมื่อรันถึง Step สุดท้าย**:
    - `POST /api/batch/complete`

สรุป (Conclusion)

หน้า **x61-MixingControl** ถูกออกแบบให้เป็น "**HMI อัจฉริยะ**" ไม่ใช่แค่หน้าจอมอนิเตอร์ แต่ทำตัวหน้าผู้ป้อนคำสั่งให้ PLC (S7-1200) ทำงานทีละคำสั่ง (Micro-management) ข้อดีคือลดภาระ Memory ของ PLC และช่วยให้ฝั่ง Software เก็บ Snapshot ข้อมูลทุก Step ได้ละเอียดที่สุด

บทที่ 4: ภาพรวมการควบคุมกระบวนการ

🏭 Process Control Concept — xMixing Production

1. Process Flow Overview

จากการวิเคราะห์ SKU ทั้ง 16 ตัวจริงจากฐานข้อมูล พบรูปแบบการผลิตดังนี้:

```
graph TD
    subgraph "Phase A - Auto Batching"
        A1[A1010<br>LS/Water/Sugar<br>Batching 40°C]
        A2[A1020<br>CMC/Emulsifier<br>+ High Shear 60°C]
    end

    subgraph "Phase D - Dissolve & Mix"
        D1[D1010<br>ละลาย Ingredient<br>60-86°C]
        D3[D1030<br>เติมสารเคมี<br>Salt/Acid/Sweetener]
    end

    subgraph "Phase x - Transfer & Control"
        X1[x1010<br>Heating Up<br>86-92°C]
        X2[x1020<br>Holding Time<br>88°C x 300-900s]
        X3[x1030<br>Cooldown<br>75-78°C<br>📊 Record Brix/pH]
        X4[x1040<br>Final Cooling<br>40-43°C<br>📊 Record Brix/pH*CTW]
    end

    A1 --> A2
    A2 --> D1
    D1 --> D3
    D3 --> X1
    X1 --> X2
    X2 --> X3
    X3 --> X4
```

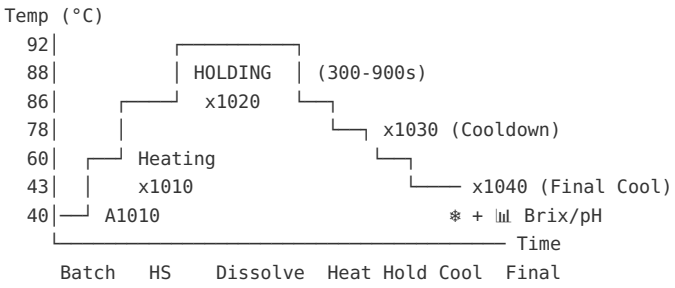
```
    style A1 fill:#bbdefb
    style A2 fill:#bbdefb
    style D1 fill:#c8e6c9
    style D3 fill:#c8e6c9
    style X1 fill:#ffe0b2
    style X2 fill:#ffccbc
    style X3 fill:#e1bee7
    style X4 fill:#b2ebf2
```

---

2. Temperature Zones (5 โซน)

Zone	Phase ID	อุณหภูมิ	Range	ระยะเวลา	วัตถุประสงค์
Mixing	A1010	**40°C**	-	ไม่กำหนด	ชั่งตวงวัตถุดิบหลัก (LS, Water, Sugar)
⚡ **High Shear**	A1020	**60°C**	-	120-180s	ผสม CMC, Emulsifier + High Shear 800-1700 RPM
🔴 **Heating**	x1010	**86-92°C**	85-95°C	ไม่กำหนด	ให้ความร้อนถึงจุดพาสเจอร์ไรส์
Holding	x1020	**88°C**	86-90°C	**300-900s**	คงอุณหภูมิฆ่าเชื้อ (Pasteurization)
❄ **Cooldown**	x1030→x1040	**78→43°C**	40-80°C	ไม่กำหนด	ลดอุณหภูมิ + เติม Flavour + QC Check

Temperature Profile Chart:



---

3. Motor Control (2 ระบบ)

3.1 Agitator (ใบกวนถังหลัก)

สถานะ	RPM	ใช้ในขั้นตอน
Low Speed	30 RPM	D1010, D1030 — ช่วงเติมวัตถุดิบละลาย
Normal Speed	60 RPM	A1020 — ช่วง High Shear ทำงาน
High Speed	80 RPM	A1010, x1010-x1040 — ช่วง Batch/Heat/Hold/Cool

3.2 High Shear (เครื่องบดละเอียด)

สถานะ	RPM	ใช้เมื่อ
OFF	0	ส่วนใหญ่ของกระบวนการ
Medium	800	เติมของเหนียว (Emulsifier, Textaid)
High	1700	เติม CMC, สารทำให้ข้น

[!IMPORTANT] High Shear ทำงานเฉพาะ Phase **A1020** เท่านั้น! และมี **timer control** (120-180 วินาที)

---

4. Recording & QC Checkpoints

4.1 ค่าที่ต้อง Record ตลอด (ทุก Step)

ค่า	Field	บันทึก	ประเภท
🌡 **อุณหภูมิ**	`qc_temp`	**ทุก Step**	Continuous
♨ **Steam Pressure**	`record_steam_pressure`	**ทุก Step**	Continuous

4.2 ค่าที่ Record เฉพาะบาง Step (QC Checkpoint)

ค่า	Field	เมื่อไหร่	Phase ที่พบ
❄ **CTW** (Cooling Tower Water)	`record_ctw`	ช่วง **Final Cooling** เท่านั้น	**x1040**
📊 **Brix**	`operation_brix_record`	ช่วง **Cooldown + Final**	**x1030, x1040**
pH	`operation_ph_record`	ช่วง **Cooldown + Final**	**x1030, x1040**

4.3 QC Checkpoint Summary (จาก Blue Lemon Freshy 32 steps)

```
Step 1-23: ไม่มี Brix/pH record → เป็นขั้นตอนผลิตปกติ
Step 24 (x1030 Cooldown 78°C):  📊 Brix    pH   ← QC Check #1
Step 25-31: ไม่มี
Step 32 (x1040 Final 43°C):     📊 Brix    pH  ❄ CTW  ← QC Check #2 (Final)
```

---

5. Data Flow: Recipe → PLC → Record

```
sequenceDiagram
    participant DB as Central DB
    participant HMI as x31 Dashboard
    participant PLC as PLC Controller
    participant REC as Production Record DB

    Note over HMI: Operator clicks START
    HMI->>PLC: 📥 Download Recipe (ALL steps)

    loop Each Step
        PLC->>PLC: Execute step (dosing/heat/mix)
        PLC->>HMI:    Telemetry (every 1s)<br>{temp, rpm, weight, timer}
```

```
    alt Step has record flag
        PLC->>HMI:    Record Request<br>{actual_temp, actual_weight, actual_brix, actual_ph}
        HMI->>REC:    Save Production Record
    end

    alt QC Checkpoint (x1030/x1040)
        PLC->>HMI:    QC Data<br>{brix_actual, ph_actual, ctw}
        HMI->>HMI: Compare vs SP (Setpoint)
        HMI->>REC:    Save QC Record
        Note over HMI:    PASS or x FAIL alert
    end
end

PLC->>HMI:    Batch Complete
HMI->>REC:    Save Batch Summary
```

---

6. Production Record Schema

6.1 Per-Step Record (บันทึกทุก Step)


```json
{
  "batch_id": "P260411-02-01-001",
  "step_no": 10,
  "phase": "p0010",
  "phase_id": "A1010",
  "re_code": "LS in Line",

  "setpoint": {
    "require": 371.71,
    "temperature": 40.0,
    "agitator_rpm": 80.0,
    "high_shear_rpm": 0
  },

  "actual": {
    "weight": 371.85,
    "temperature": 40.2,
    "agitator_rpm": 79.5,
    "high_shear_rpm": 0,
    "steam_pressure": 2.5,
    "elapsed_time": 45
  },

  "tolerance": {
    "weight_ok": true,
    "temp_ok": true
  },

  "timestamp_start": "2026-04-15T10:00:00",
  "timestamp_end": "2026-04-15T10:00:45",
  "recorded_by": "PLC"
}
```


6.2 QC Checkpoint Record (เฉพาะ x1030/x1040)


```json
{
  "batch_id": "P260411-02-01-001",
```

```
  "checkpoint": "COOLDOWN",
  "phase_id": "x1030",

  "setpoint": {
    "temperature": 78.0,
    "temp_range": [75.0, 80.0],
    "brix_sp": "14.5",
    "ph_sp": "3.8"
  },

  "actual": {
    "temperature": 77.5,
    "brix": 14.3,
    "ph": 3.82,
    "ctw_temp": null,
    "steam_pressure": 2.1
  },

  "result": "PASS",
  "operator": "admin",
  "timestamp": "2026-04-15T10:15:30",
  "notes": ""
}
```


---

7. PLC Register Mapping Concept

Setpoint Registers (HMI → PLC) — Written per step

Register	Type	Description
DB100.0	INT	Step Number
DB100.2	INT	Action Code
DB100.4	REAL	Target Weight (kg)
DB100.8	REAL	Temp Setpoint (°C)
DB100.12	REAL	Temp Low Limit
DB100.16	REAL	Temp High Limit
DB100.20	REAL	Agitator RPM SP
DB100.24	REAL	High Shear RPM SP
DB100.28	INT	Step Time (s)
DB100.30	REAL	Low Tolerance
DB100.34	REAL	High Tolerance

Actual/Telemetry Registers (PLC → HMI) — Read every 1s

Register	Type	Description
DB200.0	INT	Current Step No
DB200.2	INT	Step Timer (s)
DB200.4	REAL	Mixing Tank Temp (°C)
DB200.8	REAL	Mixing Tank Weight (kg)

DB200.12 REAL Agitator RPM Actual

DB200.16 REAL High Shear RPM Actual

DB200.20 REAL Hopper Weight (kg)

DB200.24 REAL Steam Pressure

DB200.28 REAL CTW Temp

DB200.32 INT Watchdog

DB200.34 INT Status (0=IDLE, 1=RUN, 2=PAUSE, 3=DONE, 9=ERROR)

QC Registers (Operator Input or PLC Sensor)

Register	Type	Description
DB300.0	REAL	Actual Brix
DB300.4	REAL	Actual pH
DB300.8	REAL	Actual CTW Temp
DB300.12	BOOL	QC Record Trigger
DB300.13	BOOL	QC Pass/Fail

---

8. Implementation Plan

Phase 1: Recipe Download (Done)

- ☒ Send ALL recipe to PLC via MQTT
- ☒ PLC Simulator handles recipe
- ☒ Download button + ACK

Phase 2: Step-by-Step Telemetry (Current)

- ☐ PLC simulator uses recipe SP for realistic telemetry
- ☐ Dashboard highlights active step with actual vs SP
- ☐ Step auto-advance based on step_time

Phase 3: Production Records

- ☐ Create `production_records` table in DB
- ☐ API endpoint to save per-step records
- ☐ Auto-save actual values when step completes

Phase 4: QC Checkpoints

- ☐ Detect QC steps (brix/ph/ctw flags)
- ☐ Show QC input dialog at x1030/x1040
- ☐ Compare actual vs SP → PASS/FAIL
- ☐ Save QC records to DB

Phase 5: Batch Report

- [ ] Generate batch production report
- [ ] Include all step records + QC results
- [ ] Export as PDF for QA sign-off

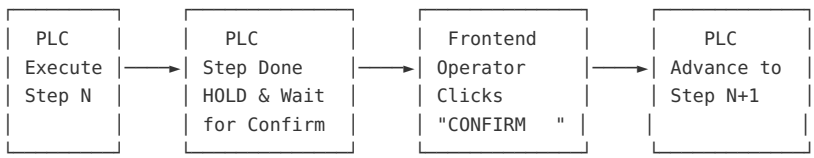
บทที่ 5: การยืนยันขั้นตอน

Step Confirmation Concept — Operator Must Confirm Every Step

The Problem

In the current “full auto” design, the PLC auto-advances after timer/condition is met. But in real food manufacturing, the **operator must visually verify** each step is actually done before moving forward (e.g., “Is the sugar fully dissolved?”, “Did the temperature reach 83°C?”).

The New Workflow: “Execute → Hold → Confirm → Advance”



PLC State Machine (Updated)

```
State 0: IDLE          → Waiting for recipe download + Start command
State 1: EXECUTING     → Running current step (timer counting, outputs ON)
State 2: WAIT_CONFIRM  → Step finished, holding outputs, waiting for operator
State 3: PAUSED        → Operator paused the batch
State 4: DONE          → All steps complete
State 9: ERROR         → Fault condition
```

Key Change: After every step completes, PLC goes to **State 2 (WAIT_CONFIRM)** instead of directly advancing. The agitator/temperature HOLD at current setpoint while waiting, so the product is safe.

Data Block Updates (DB1780)

Add these fields to the header:

```
// ── Confirmation Handshake ──
Step_Done          : Bool;    // PLC sets TRUE when step execution is finished
```

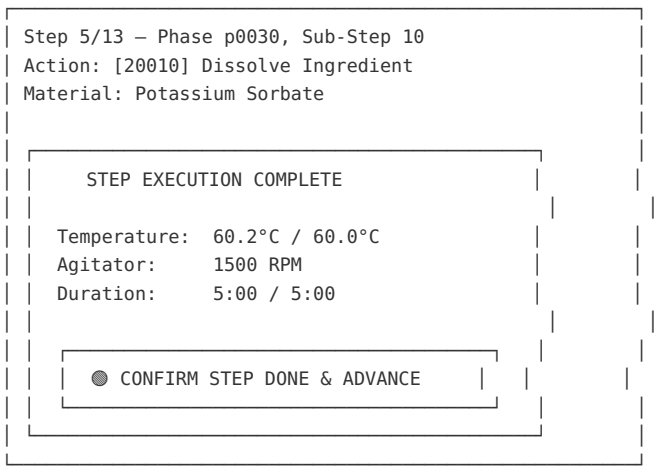
```
Confirm_Step       : Bool;    // PC sets TRUE to acknowledge and advance
Confirm_Phase_ID   : String[10];  // PC echoes back which phase it's confirming
Confirm_Step_ID    : Int;         // PC echoes back which step it's confirming
```

PLC Logic Change (in FB1780):

```
// After step timer/condition is met:
IF step_finished THEN
    "DB_RecipeData".Step_Done := TRUE;
    "DB_RecipeData".PLC_State := 2;  // WAIT_CONFIRM
    // DO NOT advance — hold current outputs
    // Agitator keeps running, temperature maintains
END_IF;

// When PC confirms:
IF "DB_RecipeData".Confirm_Step AND "DB_RecipeData".PLC_State = 2 THEN
    // Verify the confirmation matches current step (safety!)
    IF "DB_RecipeData".Confirm_Phase_ID = current_phase
       AND "DB_RecipeData".Confirm_Step_ID = current_sub_step THEN

        // Reset flags
        "DB_RecipeData".Step_Done := FALSE;
        "DB_RecipeData".Confirm_Step := FALSE;

        // ADVANCE to next step
        advance_to_next_step();
        "DB_RecipeData".PLC_State := 1;  // Back to EXECUTING
    END_IF;
END_IF;
```

Frontend Changes (x62-MixingControlV2.vue)

What the Operator Sees:



Button Behavior:

1. **Button is DISABLED** while `PLC_State = 1` (Executing) — the step is still running

2. **Button turns GREEN** when `PLC_State = 2` (Wait_Confirm) — step is done, waiting for operator
3. **Operator clicks** → Frontend publishes MQTT:

```
{
  "Confirm_Step": true,
  "Confirm_Phase_ID": "p0030",
  "Confirm_Step_ID": 10,
  "timestamp": "2026-05-10T17:55:00"
}
```

4. PLC receives → validates → advances → `PLC_State` goes back to `1`
5. Frontend sees the state change and the next step starts highlighting

---

MQTT Message Flow

Step Execution (PLC → Frontend):

```
Topic: mixing/plant/1/status
{
  "PLC_State": 1,           ← Executing
  "Phase_ID": "p0030",
  "Step_ID": 10,
  "Step_Done": false,
  "Step_Timer": 245          ← Countdown
}
```

Step Ready for Confirm (PLC → Frontend):

```
Topic: mixing/plant/1/status
{
  "PLC_State": 2,           ← Wait Confirm
  "Phase_ID": "p0030",
  "Step_ID": 10,
  "Step_Done": true,         ← Step finished!
  "Step_Timer": 0
}
```

Operator Confirms (Frontend → PLC):

```
Topic: mixing/plant/1/step_confirm
{
  "Confirm_Step": true,
  "Confirm_Phase_ID": "p0030",
  "Confirm_Step_ID": 10
}
```

PLC Advances (PLC → Frontend):

```
Topic: mixing/plant/1/status
{
  "PLC_State": 1,           ← Back to Executing
  "Phase_ID": "p0030",
```

```
  "Step_ID": 20,            ← Next step!
  "Step_Done": false,
  "Step_Timer": 300
}
```

Safety Rules

Rule	Description
Phase/Step Echo	Frontend MUST echo back which Phase+Step it's confirming. PLC rejects if mismatch.
Double-click Guard	Frontend disables the confirm button for 2 seconds after click to prevent double-advance.
Heartbeat Still Active	If PC_Heartbeat is lost while in WAIT_CONFIRM, PLC holds position indefinitely (safe state).
Timeout Warning	If WAIT_CONFIRM exceeds 10 minutes, PLC can trigger a warning alarm (but does NOT auto-advance).
Manual Steps	For ActionCode 21010 (Manual Add), the confirm button serves double duty: operator scans the ingredient AND confirms.

Summary of Changes Required

Layer	What to Change
PLC DB1780	Add `Step_Done`, `Confirm_Step`, `Confirm_Phase_ID`, `Confirm_Step_ID` fields
PLC FB1780	After step done → set State=2, hold outputs. On Confirm → validate → advance
Node-RED	Add S7-Out node to write `Confirm_Step` + IDs when MQTT confirm arrives
Frontend	Add "Confirm Step Done" button that enables only when `PLC_State = 2`
Mock Simulator	Update to wait for confirm MQTT message before advancing

---

บทที่ 6: การกู้คืนระบบ PLC

PLC & PC Concept and Recovery Workflow

Architecture for seamless recovery from App or PC shutdown during mixing execution.

To handle scenarios where the App or Computer shuts down abruptly while maintaining the ability to seamlessly recover, the best approach is the **"PLC as Executor, PC as Supervisor"** concept.

Once the production starts, the PLC must not rely on the PC to feed it instructions step-by-step. Instead, the PLC holds the recipe and manages the state machine, while the PC just watches and logs.

1. The Concept (PLC as State Master)

- **Pre-load the Recipe:** The entire recipe (or up to 50 steps) is downloaded into the PLC's `DB_Recipe` *before* the batch officially starts.
- **PLC Manages the Sequence:** The PLC uses an internal variable (e.g., `Current_Step_Idx`) to track where it is. It executes Step 1, finishes it, and automatically moves to Step 2 without asking the PC.
- **PC is just a Viewer/Logger:** The App continuously polls the PLC to read `Current_Step_Idx` and logs the actual values (actual temp, time taken, etc.) to the database.
- **Resume Capability:** Because the PLC knows the `Batch_ID` and the `Current_Step_Idx`, if the PC reboots, the PC simply asks the PLC: *"What Batch are you running, and what step are you on?"* and immediately syncs its UI back to reality.

---

2. Required Updates to your Data Block Header

To make recovery work, you need to add a few synchronization variables to your `DB_Recipe` header:

```
// Header Information
Batch_ID         : STRING[20];   // Critical: Links PLC memory to your Database Batc
SKU_ID           : STRING[20];
Total_Steps      : INT;
Current_Step_Idx : INT;          // PLC updates this as it moves forward

// Handshake & Status
Command_Start    : BOOL;         // PC tells PLC to Start
Command_Pause    : BOOL;         // PC tells PLC to Pause
PC_Heartbeat     : BOOL;         // Toggles every 1s. If PLC doesn't see toggle, PC
dead.
PLC_State        : INT;          // 0=Idle, 1=Running, 2=Waiting for Manual, 3=Pause
4=Done
```

---

3. Normal Workflow (No Crashes)

1. **Download:** Operator selects a Batch. The App writes the `Batch_ID` and the array of up to 50 steps to the PLC.
2. **Start:** App sends `Command_Start = TRUE`.
3. **Execution:** PLC reads `Steps[Current_Step_Idx]`, sets Agitator, High Shear, and Temp. It waits for the `Step_Time` to finish.
4. **Transition:** When the time is up, PLC increments `Current_Step_Idx += 1` and executes the next step automatically.
5. **Logging:** The App notices `Current_Step_Idx` changed, so it records the completion of the previous step into your `mixing_batch_step_log` table.

---

4. Recovery Workflow (App / Computer Crashes & Restarts)

Scenario: The PC dies during Step 15.

1. **PLC Reaction (PC_Heartbeat fails):**
    - The PLC detects the `PC_Heartbeat` stopped toggling.

    - **If the step is automatic** (e.g., mixing for 5 minutes), the PLC *continues* running the step to avoid ruining the product.
    - **If the step requires manual scanning** (like Action 21010 “Manual Add”), the PLC safely goes into a `Waiting` state (keeping agitator on, but timer paused) because the operator needs the PC scanner to continue.
2. **PC Reboots & App Starts:**
    - The App connects to the PLC and reads the `Batch_ID` and `Current_Step_Idx`.
    - **App Logic:** “Ah, the PLC says it is running Batch `B20260510-01` at Step `15`.”
3. **Syncing the Database:**
    - The App checks its database for `B20260510-01`.
    - It sees the batch status is still `In Progress`.
    - It sees logs up to Step 14. It knows Step 15 is currently active.
4. **Resume UI:**
    - The App instantly routes the operator to the Mixing Control screen for that Batch, highlights Step 15 as “Running”, and resumes polling data. **Recovery Complete without any operator input.**

---

5. PLC Communication Protocol: The Best Choice

Since your architecture already utilizes **Node-RED** and **RabbitMQ**, the absolute most robust way to communicate with the PLC is to decouple the PC from the PLC physically using Node-RED as middleware.

Option A: S7 Protocol via Node-RED (Recommended for Siemens S7-1200/1500) * Use the `node-red-contrib-s7` node. * **Why:** It natively reads/writes to the Data Block (DB) based on absolute offsets (e.g., `DB1,X0.0`) without requiring any special code or licenses on the PLC side. It is extremely fast and lightweight.

Option B: OPC UA (Recommended if PLC supports it natively) * Use `node-red-contrib-opcua`. * **Why:** Tag-based and highly standardized. If the PLC tags move, OPC UA doesn’t break.

The Ideal Communication Flow: 1. **Polling:** Node-RED connects to the PLC and polls the `DB_Recipe` header every 500ms. 2. **Publishing:** If `Current_Step_Idx` or `PLC_State` changes, Node-RED publishes a JSON message to a **RabbitMQ** topic (e.g., `plc.mixing.state`). 3. **Consuming:** FastAPI listens to RabbitMQ, logs the step completion securely into MySQL (`mixing_batch_step_log`), and broadcasts the live state to Nuxt.

---

6. Required Changes to Existing Applications

Backend (FastAPI) Updates

1. **Remove State-Machine Logic:** Remove any Python logic that says *“Wait 5 minutes then trigger the next step”*. FastAPI must become stateless; it only reacts to events emitted by the PLC via RabbitMQ.
2. **Add Download Endpoint:** Create a `POST /api/batch/{id}/download-to-plc` endpoint. This query formats the 50-step array and sends it to Node-RED to write to the PLC `DB_Recipe`.
3. **RabbitMQ Consumer Worker:** Create a background worker in FastAPI to consume PLC state changes from RabbitMQ and update the MySQL database instantly.

Frontend (Nuxt) Updates

1. **Make UI Stateless:** The UI must stop driving the process. Remove the manual “Next Step” buttons for any automated mixing/heating steps.
2. **WebSocket / SSE Integration:** The frontend should connect to FastAPI via WebSocket. When the PLC moves to Step 15, the frontend instantly receives the event and visually highlights Step 15 in the UI.
3. **Recovery on Mount (`onMounted`):** When the operator navigates to the Mixing Control page, the code should immediately fetch the current `PLC_State` and `Current_Step_Idx`. If the batch is already running, the UI simply jumps to that step and resumes monitoring.
4. **Heartbeat Signal:** The Nuxt frontend (or FastAPI) must implement a loop that toggles the `PC_Heartbeat` bit in the PLC every 1-2 seconds so the PLC knows the PC is alive.

บทที่ 7: Hardware Datasheet: PLC S7-1200 DB

S7-1200 PLC Data Block & Interaction Design

Method: Step-by-Step Handshake (via Node-RED & MQTT)

This document defines the data structures (Data Blocks) used for real-time communication between the HMI/App and the Siemens S7-1200 PLC.

📥 DB100: Step Command (App → PLC)

Role: Receives recipe parameters and execution commands for the current step. **Access:** App writes (via MQTT/Bridge), PLC reads.

Name	Data Type	Description	Example
`Watch_Doc`	Int	Watchdog timer/counter from App	1234
`Plan_ID`	String[20]	Production Plan Identifier	‘PLAN-2024-001’
`Batch_ID`	String[20]	Current Production Batch	‘P260411-02-01’
`SKU_Name`	String[30]	Product name or SKU identifier	‘Strawberry_Jam_01’
`Phase_ID`	String[10]	Phase identifier code	‘A1010’
`Step_ID`	Int	Unique Step ID	501
`Step_Time_SP`	Int	Target Dwell/Mixing time (seconds)	600
`Step_Status`	Int	0=Pending, 1=Active, 2=Complete	1
`Material_ID`	String[20]	Specific Raw Material ID	‘MAT-099’
`Re_Code_ID`	String[20]	Ingredient or Recipe Code	‘RM-SUG-01’
`Req_Qty`	Real	Required Quantity for step	250.5
`TT_SP`	Array[0..16] of Real	Temperature Profile Setpoints	[60.0, 65.0, …]
`Agitator_Speed`	Real	Target Agitator Speed (RPM)	30.0

`High_Shear_SP`	Real	Target High Shear Speed (RPM)	1500.0
`PH_Target`	Real	Target pH level	4.5
`Brix_Target`	Real	Target Brix value	12.5
`HMI_Command`	Int	0=IDLE, 1=START, 2=PAUSE, 3=ABORT	1
`Cmd_NewStep`	Bool	Trigger Flag to start the step	TRUE

📥 DB200: Telemetry (PLC → App)

Role: Live monitoring and feedback from the physical hardware. **Access:** PLC writes, App reads (polled or streamed every 1s).

Name	Data Type	Description	Example
`Watchdog`	Int	Heartbeat signal from PLC	32767
`PLC_State`	Int	0=Ready, 1=Run, 2=Hold, 9=Error	1
`Current_Step`	Int	Which step the PLC is actively executing	5
`Step_Timer`	Int	Elapsed time in current step (seconds)	124
`Step_Status_Act`	Int	Current Step Phase (e.g., 0=Init, 1=Dosing, 2=Mixing)	2
`MixTank_Temp`	Real	Live Tank Temperature (°C)	59.5
`MixTank_Weight`	Real	Live Tank Weight (kg)	249.8
`Agitator_Act`	Real	Live Agitator Speed (RPM)	30.1
`HighShear_Act`	Real	Live High Shear Speed (RPM)	1498.2
`PH_Actual`	Real	Live pH Sensor Reading	4.52
`Brix_Actual`	Real	Live Brix Sensor Reading	12.4
`Hopper_Weight`	Real	Live Sub-Hopper Weight (kg)	0.0

Interaction Flow (Handshake)

The communication follows a strict handshake protocol to ensure safety in manual and automatic modes.

```
sequenceDiagram
    participant App as Web Application
    participant NR as Node-RED / Bridge
    participant DB100 as PLC DB100 (CMD)
    participant DB200 as PLC DB200 (STATUS)

    Note over App,DB200: Step Execution Flow

    App->>NR: topic: plc/command (Step Data)
    NR->>DB100: Write All Params + Cmd_NewStep = 1

    Note over DB100: PLC Detects NewStep Trigger
    DB100->>DB100: Logic Starts Step
    DB100->>DB100: PLC Reset Cmd_NewStep = 0 (ACK)
```

```
loop Every 1s
    DB200-->>NR: Current_Step, Temp, Timer
    NR-->>App: UI Update (Live Dashboard)
end

Note over DB200: PLC Finished Step
DB200->>App: Step_Status_Act = Complete
```

🛠 Implementation Guidance

1. **Watchdog Logic**: Both the App (DB100) and PLC (DB200) increment their watchdog values. If a side detects that the other side's value hasn't changed for >5 seconds, it should trigger a **Communication Error** and safe-stop any active motor/pump.
2. **String Handling**: S7 strings use the first two bytes for length metadata. The App serialization must account for this (2 bytes + actual string data).
3. **Trigger Reset**: The PLC should reset `Cmd_NewStep` to `FALSE` as soon as it acknowledges the new parameters, serving as a handshake confirmation back to the App.

บทที่ 8: Hardware Datasheet: Memory Map

DB1511 Memory Map (S7-1200 Non-Optimized)

This document maps the exact byte offsets for **DB1511 (DB_FULL_RECIPE)**. This is extremely important for writing your `python-snap7` `serialize()` function, as you must pack the binary payload to match these exact byte locations.

> **Important:** This assumes the DB is set to `S7_Optimized_Access := 'FALSE'` (Standard Access) so offsets are deterministic.

📦 1. `type_FullRecipe` (Header & Array)

This is the top-level structure of DB1511. The header takes up the first 52 bytes, followed by the array of 128 steps.

Offset	Name	Data Type	Size (Bytes)	Description
`+0.0`	`Batch_ID`	String[20]	22	Batch Code (e.g. "P260411-02"). S7 strings have 2 header bytes.
`+22.0`	`SKU_ID`	String[20]	22	Recipe/SKU Code
`+44.0`	`HMI_Command`	Int	2	0=IDLE, 1=START, 2=PAUSE, 3=ABORT, 9=RESET
`+46.0`	`Total_Steps`	Int	2	How many steps are actually populated

				(1-128)
+48.0	Active_Step	Int	2	PLC Pointer — current Seq number (1-128)
+50.0	Cmd_LoadRecipe	Bool	1 (bit 0)	Trigger bit to tell PLC a new array is ready
padding	*(Padding)*	*(Byte)*	1	S7 aligns Arrays to even byte boundaries
+52.0	Steps[1]	type_RecipeStep	78	First step (See Table 2)
+130.0	Steps[2]	type_RecipeStep	78	Second step
+208.0	Steps[3]	type_RecipeStep	78	Third step
…	…	…	78	… repeats …
+9958.0	Steps[128]	type_RecipeStep	78	Last step

*(Total Size of DB1511: **10,036 bytes** ≈ 10 KB)*

🛠 2. `type_RecipeStep` (Individual Step Structure)

Each element in the `Steps[1..128]` array is exactly **78 bytes** long.

Relative Offset	Name	Data Type	Size (Bytes)	Description
+0.0	Seq	Int	2	Flat sequence number (1, 2, 3, … 128)
+2.0	Phase_No	Int	2	Phase number (10, 20, 30…)
+4.0	Sub_Step	Int	2	Step within phase (10, 20, 30…)
+6.0	Action_Code	String[10]	12	Action code (e.g. “x10010”)
+18.0	Phase_ID	String[10]	12	Phase ID (e.g. “p0010”)
+30.0	Re_Code	String[20]	22	Ingredient / Material code
+52.0	Target_Weight	Real	4	Target dosing weight (kg)
+56.0	Temp_SP	Real	4	Temperature setpoint (°C)
+60.0	Temp_Low	Real	4	Min temperature limit (°C)
+64.0	Temp_High	Real	4	Max temperature limit (°C)

+68.0	Agitator_SP	Real	4	Agitator speed (RPM)
+72.0	HighShear_SP	Real	4	High shear speed (RPM)
+76.0	Step_Time	Int	2	Hold time (Seconds)

*(Total Size per Step: **78 bytes**)*

🏷 3. Example: Cafe Amazon (S77S743200) — Flattened into Array

This is how the database recipe for SKU “Cafe Amazon” would be flattened into the `Steps[1..128]` array:

Seq	Phase_No	Sub_Step	Action_Code	Phase_ID	Ingredient	Target (kg)
1	10	10	x10010	p0010	LS in Line	371.71
2	10	20	x10020	p0010	RO-Water	372.87
3	10	30	x10030	p0010	White Sugar W150	250.0
4	20	10	x20010	p0020	*(empty)*	0.0
5	30	10	x30010	p0030	Potassium Sorbate	0.8
6	30	20	x30020	p0030	Sodium Benzoate	0.2
7	30	30	x30030	p0030	Malic Acid	0.115
8	30	40	x30040	p0030	Caramel Colour III	0.2
9	30	50	x30050	p0030	RO-Water	4.0
10	40	10	x40010	p0040	*(empty)*	0.0
11	45	10	x45010	p0045	Sugar Flavour SG-01	0.11
12	50	10	x50010	p0050	*(empty)*	0.0
13	60	10	x60010	p0060	*(empty)*	0.0
14-128	0	0			*(unused — Seq=0)*	0.0

> The PLC reads `Total_Steps = 13`, so it knows to stop after Seq 13 and ignore Steps[14] to Steps[128].

Python Serialization Tip

When you write the `.serialize()` function in `plc_interface.py`:

```python
import struct

def pack_s7_string(s: str, max_len: int) -> bytes:
    s_bytes = s.encode('ascii')[:max_len]
    return struct.pack('BB', max_len, len(s_bytes)) + s_bytes.ljust(max_len, b'\x00')
```

```
# Pack Header (52 bytes)
header = b""
header += pack_s7_string(batch_id, 20)       # +0
header += pack_s7_string(sku_id, 20)          # +22
header += struct.pack('>h', hmi_command)      # +44
header += struct.pack('>h', total_steps)      # +46
header += struct.pack('>h', active_step)      # +48
header += struct.pack('?', cmd_load_recipe)   # +50
header += b'\x00'                             # +51 padding

# Pack Each Step (78 bytes each)
for step in steps:
    header += struct.pack('>hhh', step.seq, step.phase_no, step.sub_step)  # +0,2,4
    header += pack_s7_string(step.action_code, 10)   # +6
    header += pack_s7_string(step.phase_id, 10)      # +18
    header += pack_s7_string(step.re_code, 20)        # +30
    header += struct.pack('>ffffff',
        step.target_weight, step.temp_sp, step.temp_low,
        step.temp_high, step.agitator_sp, step.highshear_sp)  # +52..+76
    header += struct.pack('>h', step.step_time)       # +76

# Total payload = 52 + (128 * 78) = 10,036 bytes
plc.db_write(db_number=1511, start=0, data=header)
```

บทที่ 9: สถาปัตยกรรม PLC Recipe

📊 SKU Recipe Analysis — All 16 Active SKUs

1. Full SKU Breakdown

SKU ID	Product Name	Steps	Phases	Max Steps/Phase	Avg Steps/Phase
S77S743200	Cafe Amazon	**13**	7	5	1.9
SFMU5USN00	Fresh Mint Senorita 750ml	**15**	8	4	1.9
S7GCA4SNA0	Japanese Melon Senorita 1900ml	**17**	8	8	2.1
S7GU5USN00	Japanese Melon Senorita 750ml	**17**	8	8	2.1
S7KCY4SNA0	Coconut Senorita 1900ml	**17**	9	4	1.9
S7KU5USN00	Coconut Senorita 750ml	**17**	9	4	1.9
S77CY4SN00	Classic Caramel Senorita	**18**	9	6	2.0
SFLDBUSS00	Rose Senorita Signature 720ml	**24**	10	7	2.4
S7EFRU4200-	Strawberry	**30**	15	5	2.0

1	Freshy 710ml x12				
S7EFSU4200	Strawberry Freshy 710ml x3	**30**	15	5	2.0
S7HFRU4200	Orange Freshy 710ml x12	**31**	12	**15**	2.6
S7CFSU4200	Blue Lemon Freshy 710ml x3	**32**	17	5	1.9
S7DFRU4200	Lychee Freshy 710ml x12	**34**	16	5	2.1
S7DFSU4200	Lychee Freshy 710ml x3	**34**	16	5	2.1
S7YFRU4200	Green Apple Freshy 710ml x12	**34**	18	5	1.9
SFGFSU4200	Blue Hawaii Freshy 710ml x3	**35**	17	5	2.1

---

2. Statistical Summary

```
┌──────────────────────────────────────────────┐
│          16 SKUs Analyzed                    │
├──────────────────────┬──────┬──────┬─────────┤
│ Metric               │ Min  │ Max  │   Avg   │
├──────────────────────┼──────┼──────┼─────────┤
│ Steps per SKU        │  13  │  35  │  24.9   │
│ Phases per SKU       │   7  │  18  │  12.1   │
│ Max steps in 1 phase │   -  │  15  │   -     │
│ Avg steps per phase  │ 1.9  │ 2.6  │  2.1    │
└──────────────────────┴──────┴──────┴─────────┘
```

---

3. Phase Type Classification

พบ 3 ประเภท Phase ในทุก SKU:

Prefix	Type	หน้าที่	ตัวอย่าง
A	**Auto Batching**	ชั่งตวงวัตถุดิบอัตโนมัติ (น้ำ, น้ำตาล, LS)	A1010, A1020
D	**Dissolve / Mix**	ละลาย, ผสม, ต้ม, เติมวัตถุดิบมือ	D1010, D1030
x	**Transfer / Control**	โอนย้ายถัง, รอ, QC, จบกระบวนการ	x1010, x1020, x1030, x1040

Process Flow Pattern (พบในทุก SKU):

```
graph LR
    A[A - Auto Batching<br>ชั่งตวงอัตโนมัติ] --> X1[x - Transfer<br>โอนเข้าถังผสม]
    X1 --> D[D - Dissolve<br>ละลาย/ผสม]
    D --> X2[x - Transfer<br>โอนออก]
```

```
    X2 --> X3[x1020 - QC Check]
    X3 --> X4[x1030 - Storage]
    X4 --> X5[x1040 - Complete]
```

---

4. Payload Size Analysis

Method	Worst Case	Typical	Comment
Send ALL	35 steps × 200B = **6.8 KB**	25 × 200B = **4.9 KB**	เล็กมาก!
Send per Phase	15 steps × 200B = **2.9 KB**	2 × 200B = **0.4 KB**	เล็กเกินไป
Send per Step	1 × 200B = **0.2 KB**	-	ไม่คุ้ม overhead

> [!IMPORTANT] **ข้อมูลสำคัญ**: Payload ของสูตรที่ใหญ่ที่สุด (35 steps) มีขนาดเพียง **6.8 KB** เท่านั้น — **เล็กมากๆ** สำหรับ PLC สมัยใหม่ที่มี memory เป็น MB ขึ้นไป

---

5. 🏆 คำแนะนำ: Send ALL (Option A)

เหตุผล:

#	เหตุผล	รายละเอียด
1	**Payload เล็กมาก**	แม้ SKU ที่ซับซ้อนสุด (35 steps) ก็แค่ 6.8 KB — PLC รับได้สบายๆ
2	**Avg 2.1 steps/phase**	ถ้าส่งทีละ Phase จะต้อง handshake 12-18 ครั้ง เปลืองเกินไป
3	**PLC ทำงานอิสระ**	ไม่ต้องรอ Network ระหว่างขั้นตอน — **ปลอดภัยกว่า** สำหรับโรงงาน
4	**ง่ายต่อการ Debug**	ส่งครั้งเดียว ตรวจสอบว่าครบถ้วน แล้ว PLC วิ่งต่อเอง
5	**Network Failure Safe**	ถ้าเน็ต MQTT หลุดระหว่างผลิต PLC ยังวิ่งต่อได้

Architecture ที่แนะนำ:

```
sequenceDiagram
    participant OP as Operator
    participant HMI as x31 Dashboard
    participant MQTT as MQTT Broker
    participant PLC as PLC

    OP->>HMI: Click "Start Production"
    HMI->>HMI: Fetch all SKU steps from DB

    HMI->>MQTT: mixing/plant/1/recipe<br>{batch_id, sku_id, batch_size,<br>steps: [all 13-35
steps]}
    MQTT->>PLC: Full recipe (max 6.8 KB)
    PLC->>MQTT: mixing/plant/1/ack<br>{status: "RECIPE_LOADED", steps: 35}

    Note over HMI:   Recipe confirmed, PLC ready
    OP->>HMI: Click "START"
    HMI->>MQTT: mixing/plant/1/cmd<br>{command: "START"}
```

```
    loop PLC executes autonomously
        PLC->>MQTT: mixing/plant/1/telemetry<br>{step_no, temp, rpm, weight, watchdog}
        HMI->>HMI: Update gauges in real-time
    end

    PLC->>MQTT: mixing/plant/1/status<br>{status: "BATCH_COMPLETE"}
```

MQTT Topic Design (Final):

```
mixing/plant/{id}/
├── recipe          # HMI → PLC: Full recipe download (ALL steps)
├── cmd             # HMI → PLC: START, PAUSE, RESUME, ABORT
├── ack             # PLC → HMI: Recipe loaded / Command acknowledged
├── telemetry       # PLC → HMI: Real-time sensor data (every 1-2s)
├── status          # PLC → HMI: RUNNING, PAUSED, COMPLETE, ERROR
└── watchdog        # PLC → HMI: Heartbeat counter
```

Recipe JSON Schema:


```
        {
          "batch_id": "P260411-02-01-001",
          "sku_id": "S77S743200",
          "sku_name": "Cafe Amazon",
          "batch_size": 1200.0,
          "plant_id": 1,
          "total_steps": 13,
          "total_phases": 7,
          "timestamp": "2026-04-15T16:40:00",
          "steps": [
            {
              "step_no": 10,
              "phase": "p0010",
              "phase_id": "A1010",
              "action_code": "10030",
              "re_code": "LS in Line",
              "destination": "1010",
              "require": 371.71,
              "uom": "kg",
              "low_tol": 0.001,
              "high_tol": 0.001,
              "agitator_rpm": 80.0,
              "high_shear_rpm": 0.0,
              "temperature": 40.0,
              "temp_low": 0.0,
              "temp_high": 0.0,
              "step_time": 0,
              "step_timer_control": 0
            }
          ]
        }
```


---

6. Next Steps

- [ ] Implement recipe publish in `x61-MixingControl.vue` (Send All on Start)
- [ ] Update `plc_simulator.mjs` to receive & parse full recipe
- [ ] Add recipe ACK handling in `useMQTT.ts`

- [ ] Add step progress tracking on dashboard (current step highlight)

บทที่ 10: System Architecture

🏭 Hybrid Architecture & Full Recipe Workflow

This document outlines the **Hybrid Communication Model** (CQRS) combined with the **Full Recipe Array (DB1511)** concept for the Mixing Control application.

🏗️ 1. System Architecture Diagram

```
graph TD
    subgraph Frontend [Vue.js Tablet UI]
        UI[Vue Components]
        MQTT_Sub[useMQTT.ts]
        API_Call[fetch / axios]
    end

    subgraph Backend [Python FastAPI]
        DB[(MySQL Database)]
        API[FastAPI Routes]
        Snap7[python-snap7]
    end

    subgraph Middleware [Telemetry Bridge]
        NR[Node-RED]
        MQ[RabbitMQ / MQTT]
    end

    subgraph PLC [Siemens S7-1200]
        DB1511[DB1511: Full Recipe Array]
        DB1512[DB1512: Live Telemetry]
        DB1513[DB1513: Handshake]
    end

    %% Telemetry Flow (Fast & Visual)
    DB1512 -.->|Read| NR
    NR -.->|Publish| MQ
    MQ -.->|Subscribe| MQTT_Sub
    MQTT_Sub -.-> UI

    %% Command Flow (Strict & Transactional)
    UI ==>|1. HTTP POST /start-batch| API_Call
    API_Call ==> API
    API ==>|2. Validate & Read| DB
    API ==>|3. Write Full Array| Snap7
    Snap7 ==>|Direct S7 Write| DB1511

    %% Handshake Feedback (Database Sync)
    DB1513 -.->|Read End-of-Step| Snap7
    Snap7 ==>|4. Update Logs| DB
```

---

2. Step-by-Step App Workflow

Phase A: Starting the Batch (The Command)

1. **Operator Action:** The operator scans a barcode and clicks “Start Production” on the Vue.js tablet.
2. **API Request:** The frontend sends an HTTP `POST /api/production/start` to FastAPI.
3. **Database Assembly:** FastAPI queries `v_sku_complete` to get all 20 steps for that specific SKU.
4. **Data Packing:** FastAPI uses `plc_interface.py` to pack all 20 steps into the binary array structure required by `type_FullRecipe`.
5. **Direct PLC Write:** FastAPI uses `snap7` to write the entire binary array directly to `DB1511` and sets `Cmd_LoadRecipe = True`.
6. **Confirmation:** FastAPI responds to the frontend with `200 OK`. (If the PLC is offline, FastAPI blocks the start and returns a 500 error).

Phase B: Mixing Execution (The Telemetry)

1. **PLC Takes Over:** The PLC sees `Cmd_LoadRecipe = True`, loads the array, and begins executing `Active_Step = 1`.
2. **Telemetry Streaming:** Node-RED continuously reads `DB1512` every 500ms and publishes to MQTT.
3. **UI Updates:** The Vue.js frontend receives the MQTT payload and updates the temperature gauges, active row highlighting, and timer *instantly*.

Phase C: Step Completion (The Handshake)

1. **PLC Finishes Step:** The PLC finishes Step 1, pulses `DB1513.Step_Complete`, and instantly moves to Step 2.
2. **Python Background Worker:** A background task in FastAPI (polling `DB1513` via snap7) detects the pulse.
3. **Database Logging:** FastAPI logs the final weight, temperature, and end-time of Step 1 into the SQL database.

📦 3. Required Application Modules

To build this, you need to structure your app modules like this:

Python Backend (FastAPI)

- **`plc_interface.py` (Update Needed):**
  - Add a new Pydantic model for `DB1511_FullRecipe` and `DB1511_RecipeStep`.
  - Write the `.serialize()` method to convert the array of steps into S7 byte format.
- **`plc_service.py` (New):**
  - Use `python-snap7` to manage the TCP connection to the PLC.
  - Function: `write_recipe_to_plc(batch_id, steps_array)`
- **`worker_handshake.py` (New):**
  - An `asyncio` background loop that reads `DB1513` (Handshake) every 1 second and writes step

completion data to the SQL database.

- **`routes/production.py`:**
  - Endpoint for `POST /start-batch` that triggers the PLC write.

Node-RED (Middleware)

- **Remove DB1510 Logic:** Delete all nodes that currently write to DB1510. Node-RED is no longer responsible for sending recipes.
- **Keep DB1512 Logic:** Keep the S7-Read nodes for DB1512 and publish them to MQTT exactly as they are.

🔵 Vue.js Frontend (x3101-0110-frontEnd)

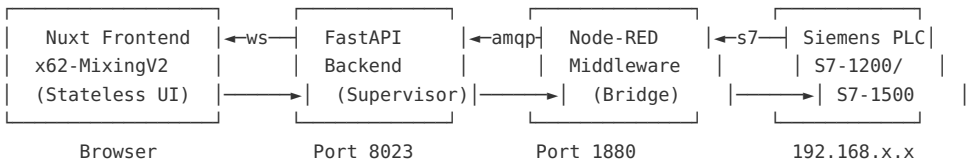
- **`x61-MixingControl.vue`:**
  - When pressing start/next, instead of publishing an MQTT message, call an Axios HTTP POST request to your FastAPI backend.
- **`useMQTT.ts`:**
  - Keep this exactly as it is! It will continue to listen to the telemetry stream to update the UI perfectly.

บทที่ 11: คู่มือติดตั้งระบบ

Production Deployment Guide — PLC Mixing Control V2

> This document explains **exactly** what needs to happen on each layer of the system to move from our tested mock environment to a real, running production line.

Architecture Overview



Data Flow: 1. **Download:** Nuxt → FastAPI → Node-RED → PLC (write recipe once) 2. **Telemetry:** PLC → Node-RED (poll 500ms) → RabbitMQ → FastAPI → Nuxt (live state) 3. **Heartbeat:** FastAPI → RabbitMQ → Node-RED → PLC (toggle bit every 1s)

PART A: PLC Code (TIA Portal)

Important: All code below is in SCL (Structured Control Language) for Siemens TIA Portal V16+. Your PLC engineer should create these in the TIA Portal project.

A1. User-Defined Types (UDTs)

Create these two UDTs first. They define the shape of every recipe step and the overall recipe.

```
// ═══════════════════════════════════════════
// UDT: "UDT_ProcessStep"
// Size: ~136 bytes per step
// ═══════════════════════════════════════════
TYPE "UDT_ProcessStep"
VERSION : 0.1
   STRUCT
      StepNo        : Int;          // Sub-step number (1, 2, 3...)
      ActionCode    : Int;          // Action code (10010=Setup, 21010=Manual Add, etc.)
      ReCode        : String[25];   // Recipe ingredient code
      SapCode       : String[20];   // SAP Material code
      Destination   : Int;          // Target vessel (0=MixTank, 1=Hopper...)
      Require       : Real;         // Required weight (kg)
      LowTol        : Real;         // Low tolerance (kg)
      HighTol       : Real;         // High tolerance (kg)
      Temperature   : Real;         // Setpoint temperature (°C)
      TempLow       : Real;         // Temp low limit
      TempHigh      : Real;         // Temp high limit
      AgitatorRPM   : Real;         // Agitator speed setpoint (RPM)
      HighShearRPM  : Real;         // High-shear mixer speed (RPM)
      StepTime      : Int;          // Step duration (seconds)
      StepTimerCtl  : Int;          // Timer control mode (0=none, 1=auto, 2=manual)
      SetupStep     : Int;          // Setup step flag
      Condition     : Int;          // Step condition code
      QcTemp        : Bool;         // Requires QC temperature check
      RecordSteam   : Bool;         // Record steam pressure
      RecordCTW     : Bool;         // Record CTW data
      BrixRecord    : Bool;         // Requires Brix recording
      PhRecord      : Bool;         // Requires pH recording
      MasterStep    : Bool;         // Is a master step (phase start)
      StepActive    : Bool;         // Step has valid data (not empty)
      BrixSP        : Real;         // Brix setpoint
      PhSP          : Real;         // pH setpoint
   END_STRUCT;
END_TYPE

// ═══════════════════════════════════════════
// UDT: "UDT_Process" (Phase)
// Each process contains up to 8 steps
// ═══════════════════════════════════════════
TYPE "UDT_Process"
VERSION : 0.1
   STRUCT
      ProcessNo     : Int;          // Phase number (10, 20, 30...)
      PhaseID       : Int;          // Phase ID from DB
      StepCount     : Int;          // Number of active steps in this phase
      ProcessActive : Bool;         // Phase has valid data
      Steps         : Array[0..7] of "UDT_ProcessStep";
   END_STRUCT;
END_TYPE
```

A2. Data Block — DB1780 “DB_RecipeData”

This is the main recipe data block that the PC writes to and the PLC reads from.

```
// ═══════════════════════════════════════════════
// DB1780: "DB_RecipeData"
// The PC writes recipe data here BEFORE starting
// ═══════════════════════════════════════════════
DATA_BLOCK "DB_RecipeData"
{ S7_Optimized_Access := 'FALSE' }
VERSION : 0.1
   STRUCT
      // ── Header (written by PC) ──
      PlanID        : String[30];
      BatchID       : String[20];
      SkuID         : String[20];
      SkuName       : String[50];
      PlantID       : Int;
      BatchSize     : Real;
      ProcessCount  : Int;          // Number of active phases

      // ── Control Flags (written by PC) ──
      RecipeReady   : Bool;         // PC sets TRUE after download complete
      StartCmd      : Bool;         // PC sets TRUE to start batch
      PauseCmd      : Bool;         // PC sets TRUE to pause
      AbortCmd      : Bool;         // PC sets TRUE to emergency stop
      PC_Heartbeat  : Bool;         // PC toggles every 1 second

      // ── Status (written by PLC) ──
      PLC_State         : Int;      // 0=Idle, 1=Running, 2=WaitManual, 3=Paused, 4=Done,
9=Error
      Current_Process   : Int;      // Current active phase index (0-31)
      Current_Step      : Int;      // Current active step index within phase (0-7)
      Current_Step_Flat : Int;      // Flat step counter across all phases (1-based)
      Step_Timer_Act    : Int;      // Remaining seconds on current step timer
      Step_Complete     : Bool;     // Pulse: PLC sets TRUE when a step finishes
      Batch_Complete    : Bool;     // TRUE when all steps are done
      Error_Code        : Int;      // 0=None, 1=Heartbeat Lost, 2=Timeout

      // ── Actuals (written by PLC from I/O) ──
      MixTank_Temp_Act  : Real;     // Actual tank temperature
      MixTank_Weight_Act: Real;     // Actual tank weight
      Agitator_Act      : Real;     // Actual agitator RPM
      HighShear_Act     : Real;     // Actual high-shear RPM
      Hopper_Weight_Act : Real;     // Actual hopper weight
      Brix_Act          : Real;     // Actual Brix reading
      PH_Act            : Real;     // Actual pH reading

      // ── Recipe Array (32 phases × 8 steps) ──
      Processes     : Array[0..31] of "UDT_Process";
   END_STRUCT;
BEGIN
END_DATA_BLOCK
```

> ⚠ **CRITICAL:** Set `S7_Optimized_Access := 'FALSE'` so Node-RED can read/write using absolute byte offsets (e.g., `DB1780,DBD100`).

A3. Function Block — FB1780 “FB_MixingSequencer”

This is the **brain** of the system. It runs in a cyclic OB (e.g., OB1 or OB35) and manages the step-by-step execution.

```
// ═══════════════════════════════════════════════════════════
// FB1780: "FB_MixingSequencer"
// Main sequencer logic — runs in OB1 every scan cycle
// ═══════════════════════════════════════════════════════════
FUNCTION_BLOCK "FB_MixingSequencer"
VERSION : 0.1

VAR_INPUT
    // Analog inputs from field instruments
    Act_MixTank_Temp   : Real;
    Act_MixTank_Weight : Real;
    Act_Agitator_RPM   : Real;
    Act_HighShear_RPM  : Real;
    Act_Hopper_Weight  : Real;
    Act_Brix           : Real;
    Act_PH             : Real;
END_VAR

VAR_OUTPUT
    // Analog outputs to field devices
    SP_Agitator_RPM    : Real;
    SP_HighShear_RPM   : Real;
    SP_Temperature     : Real;
END_VAR

VAR
    stepTimer          : TON;         // IEC Timer for step duration
    heartbeatTimer     : TON;         // Heartbeat watchdog timer
    lastHeartbeat      : Bool;        // Previous heartbeat state
    flatStepCounter    : Int;         // Running count across all phases
END_VAR

VAR_TEMP
    curProcess         : Int;
    curStep            : Int;
    stepData           : "UDT_ProcessStep";
    stepTimeSec        : Time;
END_VAR

BEGIN

// ═══ 1. UPDATE ACTUALS ═══
"DB_RecipeData".MixTank_Temp_Act   := Act_MixTank_Temp;
"DB_RecipeData".MixTank_Weight_Act := Act_MixTank_Weight;
"DB_RecipeData".Agitator_Act       := Act_Agitator_RPM;
"DB_RecipeData".HighShear_Act      := Act_HighShear_RPM;
"DB_RecipeData".Hopper_Weight_Act  := Act_Hopper_Weight;
"DB_RecipeData".Brix_Act           := Act_Brix;
"DB_RecipeData".PH_Act             := Act_PH;

// ═══ 2. HEARTBEAT WATCHDOG ═══
// If PC_Heartbeat stops toggling for 5 seconds → PC is dead
IF "DB_RecipeData".PC_Heartbeat <> #lastHeartbeat THEN
    #lastHeartbeat := "DB_RecipeData".PC_Heartbeat;
    #heartbeatTimer(IN := FALSE, PT := T#5s);  // Reset timer
END_IF;
#heartbeatTimer(IN := TRUE, PT := T#5s);

IF #heartbeatTimer.Q AND "DB_RecipeData".PLC_State = 1 THEN
    // PC lost! But don't stop the batch — just log the error
    "DB_RecipeData".Error_Code := 1;  // Heartbeat lost
    // If current step is manual (ActionCode 21010), pause
```

```
    #curProcess := "DB_RecipeData".Current_Process;
    #curStep    := "DB_RecipeData".Current_Step;
    #stepData   := "DB_RecipeData".Processes[#curProcess].Steps[#curStep];
    IF #stepData.ActionCode = 21010 THEN
        "DB_RecipeData".PLC_State := 2;  // WaitManual — safe pause
    END_IF;
    // Automatic steps (mixing, heating) CONTINUE running
END_IF;

// ═══ 3. COMMAND HANDLING ═══
// Start
IF "DB_RecipeData".StartCmd AND "DB_RecipeData".RecipeReady THEN
    IF "DB_RecipeData".PLC_State = 0 OR "DB_RecipeData".PLC_State = 3 THEN
        "DB_RecipeData".PLC_State := 1;  // Running
        "DB_RecipeData".StartCmd  := FALSE;
        "DB_RecipeData".Error_Code := 0;
        IF "DB_RecipeData".Current_Process = 0 AND "DB_RecipeData".Current_Step = 0 THEN
            #flatStepCounter := 1;
        END_IF;
    END_IF;
END_IF;

// Pause
IF "DB_RecipeData".PauseCmd THEN
    "DB_RecipeData".PLC_State := 3;  // Paused
    "DB_RecipeData".PauseCmd  := FALSE;
END_IF;

// Abort
IF "DB_RecipeData".AbortCmd THEN
    "DB_RecipeData".PLC_State     := 0;  // Idle
    "DB_RecipeData".AbortCmd      := FALSE;
    "DB_RecipeData".Current_Process := 0;
    "DB_RecipeData".Current_Step    := 0;
    #flatStepCounter := 0;
    SP_Agitator_RPM  := 0.0;
    SP_HighShear_RPM := 0.0;
    SP_Temperature   := 0.0;
END_IF;

// ═══ 4. MAIN SEQUENCER (only when Running) ═══
IF "DB_RecipeData".PLC_State = 1 THEN

    #curProcess := "DB_RecipeData".Current_Process;
    #curStep    := "DB_RecipeData".Current_Step;

    // Safety: check bounds
    IF #curProcess > 31 OR NOT "DB_RecipeData".Processes[#curProcess].ProcessActive THEN
        "DB_RecipeData".PLC_State     := 4;  // Done
        "DB_RecipeData".Batch_Complete := TRUE;
        SP_Agitator_RPM  := 0.0;
        SP_HighShear_RPM := 0.0;
        RETURN;
    END_IF;

    // Get current step data
    #stepData := "DB_RecipeData".Processes[#curProcess].Steps[#curStep];

    // ── Apply setpoints to outputs ──
    SP_Agitator_RPM  := #stepData.AgitatorRPM;
    SP_HighShear_RPM := #stepData.HighShearRPM;
    SP_Temperature   := #stepData.Temperature;
```

```
        // ── Update flat counter for UI ──
        "DB_RecipeData".Current_Step_Flat := #flatStepCounter;

        // ── MANUAL STEP (ActionCode 21010): Wait for operator ──
        IF #stepData.ActionCode = 21010 THEN
            "DB_RecipeData".PLC_State := 2;  // WaitManual
            // The PC will set StartCmd=TRUE again when operator scans/confirms
            RETURN;
        END_IF;

        // ── TIMED STEP: Run step timer ──
        IF #stepData.StepTime > 0 THEN
            #stepTimeSec := INT_TO_TIME(#stepData.StepTime * 1000);
            #stepTimer(IN := TRUE, PT := #stepTimeSec);
            "DB_RecipeData".Step_Timer_Act :=
                TIME_TO_INT(#stepTimeSec - #stepTimer.ET) / 1000;

            IF NOT #stepTimer.Q THEN
                RETURN;  // Still counting, come back next scan
            END_IF;
            // Timer finished — reset and advance
            #stepTimer(IN := FALSE, PT := #stepTimeSec);
        END_IF;

        // ── STEP COMPLETE → advance ──
        "DB_RecipeData".Step_Complete := TRUE;  // Pulse for Node-RED to catch

        // Move to next step in this phase
        IF #curStep + 1 < "DB_RecipeData".Processes[#curProcess].StepCount THEN
            "DB_RecipeData".Current_Step := #curStep + 1;
            #flatStepCounter := #flatStepCounter + 1;
        ELSE
            // Move to next phase
            "DB_RecipeData".Current_Step := 0;
            "DB_RecipeData".Current_Process := #curProcess + 1;
            #flatStepCounter := #flatStepCounter + 1;

            // Check if we've exceeded all active phases
            IF #curProcess + 1 >= "DB_RecipeData".ProcessCount THEN
                "DB_RecipeData".PLC_State     := 4;  // Done
                "DB_RecipeData".Batch_Complete := TRUE;
                SP_Agitator_RPM  := 0.0;
                SP_HighShear_RPM := 0.0;
            END_IF;
        END_IF;

    END_IF;

    END_FUNCTION_BLOCK
```

A4. Calling FB1780 in OB1

In your main program (OB1), create an instance DB and call the function block:

```
    // In OB1 (Main Program)
    "FB_MixingSequencer_DB"(
        Act_MixTank_Temp   := "AI_MixTank_TT01",      // Your analog input tag
        Act_MixTank_Weight := "AI_MixTank_WT01",
        Act_Agitator_RPM   := "AI_Agitator_Speed",
```

```
    Act_HighShear_RPM  := "AI_HighShear_Speed",
    Act_Hopper_Weight  := "AI_Hopper_WT01",
    Act_Brix           := "AI_Brix_Sensor",
    Act_PH             := "AI_PH_Sensor",
    SP_Agitator_RPM    => "AO_Agitator_SP",        // Your analog output tag
    SP_HighShear_RPM   => "AO_HighShear_SP",
    SP_Temperature     => "AO_Temperature_SP"
);
```

Note: Replace `"AI_xxx"` and `"AO_xxx"` with your actual I/O tag names from the TIA Portal hardware configuration.

PART B: Node-RED Configuration

B1. Install Required Nodes

```
cd ~/.node-red
npm install node-red-contrib-s7
npm install node-red-contrib-amqp
```

B2. S7 Connection Configuration

In Node-RED, create an S7 connection: | Setting | Value | |—————-|—————————| | Host | `192.168.1.1` (your PLC) | | Port | `102` | | Rack | `0` | | Slot | `1` | | Cycle Time | `500` ms |

B3. S7 Variable Mapping (Read from PLC → Publish to RabbitMQ)

Create an **S7-In** node that reads these DB1780 addresses every 500ms:

Variable Name	S7 Address	Type
PLC_State	`DB1780,INT18`	INT
Current_Process	`DB1780,INT20`	INT
Current_Step	`DB1780,INT22`	INT
Current_Step_Flat	`DB1780,INT24`	INT
Step_Timer_Act	`DB1780,INT26`	INT
Step_Complete	`DB1780,X28.0`	BOOL
Batch_Complete	`DB1780,X28.1`	BOOL
Error_Code	`DB1780,INT30`	INT
MixTank_Temp_Act	`DB1780,REAL32`	REAL
MixTank_Weight_Act	`DB1780,REAL36`	REAL
Agitator_Act	`DB1780,REAL40`	REAL
HighShear_Act	`DB1780,REAL44`	REAL
Hopper_Weight_Act	`DB1780,REAL48`	REAL
Brix_Act	`DB1780,REAL52`	REAL
PH_Act	`DB1780,REAL56`	REAL
BatchID	`DB1780,S34.20`	STRING

> ⚠ **Important:** The exact byte offsets above are **approximate**. After you create DB1780 in TIA Portal with `Optimized Access = FALSE`, open the DB and look at the **Offset** column. Use those exact values in Node-RED.

B4. Node-RED Flow Logic

```
[S7-In: Read PLC every 500ms]
       |
       ▼
[Function: Compare with previous state]
       | (only publish if PLC_State or Current_Step changed)
       ▼
[AMQP-Out: Publish to RabbitMQ exchange "plc.mixing.state"]
       |
       ▼
  FastAPI Consumer picks it up
```

Function Node (change detection):

```
// Store previous state in flow context
var prev = flow.get('plc_state') || {};
var cur  = msg.payload;

// Only publish if something changed
if (prev.PLC_State !== cur.PLC_State ||
    prev.Current_Step_Flat !== cur.Current_Step_Flat ||
    prev.Step_Complete !== cur.Step_Complete) {

    flow.set('plc_state', cur);
    msg.payload = {
        PLC_State: cur.PLC_State,
        Current_Process: cur.Current_Process,
        Current_Step: cur.Current_Step,
        Current_Step_Flat: cur.Current_Step_Flat,
        Step_Timer_Act: cur.Step_Timer_Act,
        Step_Complete: cur.Step_Complete,
        Batch_Complete: cur.Batch_Complete,
        BatchID: cur.BatchID,
        MixTank_Temp_Act: cur.MixTank_Temp_Act,
        MixTank_Weight_Act: cur.MixTank_Weight_Act,
        Agitator_Act: cur.Agitator_Act,
        HighShear_Act: cur.HighShear_Act,
        timestamp: new Date().toISOString()
    };
    return msg;
}
return null; // Don't publish if nothing changed
```

B5. Writing Recipe TO PLC (PC → Node-RED → PLC)

Create an **HTTP-In** node (POST `/api/plc/write-recipe`) that accepts the JSON recipe from FastAPI and writes it to DB1780 using S7-Out nodes.

Flow:

```
[HTTP-In: POST /api/plc/write-recipe]
       |
       ▼
```

```
[Function: Map JSON to S7 writes]
       |
       ▼
[S7-Out: Write header fields]
[S7-Out: Write Process[0].Steps[0..7]]
[S7-Out: Write Process[1].Steps[0..7]]
  ... (loop through all active processes)
       |
       ▼
[S7-Out: Write RecipeReady = TRUE]
       |
       ▼
[HTTP-Response: 200 OK]
```

B6. Heartbeat Relay (PC → PLC)

```
[AMQP-In: Subscribe "pc.heartbeat"]
       |
       ▼
[S7-Out: Write DB1780.PC_Heartbeat (toggle TRUE/FALSE)]
```

PART C: Production Deployment Checklist

Step 1: PLC Setup (TIA Portal)

- ☐ Create `UDT_ProcessStep` and `UDT_Process` in TIA Portal
- ☐ Create `DB1780` with `Optimized Access = FALSE`
- ☐ Create `FB1780` with the sequencer logic
- ☐ Call `FB1780` in OB1 with correct I/O mappings
- ☐ Download to PLC and verify DB1780 appears in the watch table
- ☐ Note the **exact byte offsets** from TIA Portal's DB view

Step 2: Node-RED Setup

- ☐ Install `node-red-contrib-s7` and `node-red-contrib-amqp`
- ☐ Configure S7 connection to PLC IP (test with a simple read first)
- ☐ Map the S7 variables using exact offsets from Step 1
- ☐ Create the polling flow (read every 500ms)
- ☐ Create the change-detection function node
- ☐ Create the RabbitMQ publish node
- ☐ Create the recipe-write HTTP endpoint
- ☐ Create the heartbeat relay flow
- ☐ **Test:** Read PLC_State from Node-RED debug panel

Step 3: FastAPI Backend

- ☐ Update `POST /plc/send-recipe/{batch_id}` to POST the JSON to Node-RED's HTTP endpoint instead of returning it
- ☐ Add RabbitMQ consumer to listen for `plc.mixing.state` events

- [ ] On each event, write to `mixing_batch_step_log` table
- [ ] Add WebSocket broadcast to push state to frontend
- [ ] Add heartbeat publisher (toggle `pc.heartbeat` every 1s)

Step 4: Frontend (Already Done)

- [x] `x62-MixingControlV2.vue` is stateless
- [x] Reacts to `plantData` telemetry updates
- [x] Recovery on mount works
- [x] `downloadRecipeToPlc()` calls the backend API

Step 5: Integration Test

- [ ] Load a test batch on the PLC via the UI
- [ ] Verify recipe appears in DB1780 watch table
- [ ] Click START → verify PLC_State changes to 1
- [ ] Watch steps auto-advance in the UI
- [ ] **Pull-the-plug test:** Kill FastAPI at Step 5, restart, verify UI recovers at the correct step
- [ ] **Manual step test:** Verify PLC pauses at ActionCode 21010 and waits for operator confirmation

---

PART D: Action Code Reference for PLC Logic

The PLC sequencer needs to know which steps are automatic and which require manual intervention:

Action Code	Description	PLC Behavior
`10010`	Setup / Initialize	Auto — apply setpoints, no wait
`10020`	Heating	Auto — wait for temp + time
`10030`	Mixing / Blending	Auto — run agitator for time
`10040`	Pasteurize	Auto — hold temp for time
`21010`	Manual Add to Mix Tank	**MANUAL** — PLC waits for PC
`30020`	Manual Transfer	**MANUAL** — PLC waits for PC
`40010`	QC Check / Record	**MANUAL** — requires Brix/pH

> For all `21010` and `30020` steps, the PLC sets `PLC_State = 2` (WaitManual) and holds position until the PC sends `StartCmd = TRUE` after the operator confirms.

PART E: Network Requirements

Device	IP Address	Port	Protocol
PLC	`192.168.1.1`	102	S7 (TCP)
Node-RED	`192.168.1.100`	1880	HTTP
RabbitMQ	`192.168.1.100`	5672	AMQP
FastAPI	`192.168.1.100`	8023	HTTP
Nuxt	`192.168.1.100`	3000	HTTP/WS

All middleware (Node-RED, RabbitMQ, FastAPI, Nuxt) can run on a **single industrial PC**. Only the PLC needs a separate Ethernet connection.

PART F: Safety Considerations

1. **Emergency Stop:** The PLC's hardware E-Stop circuit must be independent of this software. The `AbortCmd` is a software-level stop only.
2. **Heartbeat Timeout:** If `PC_Heartbeat` stops for 5 seconds, the PLC continues automatic steps but pauses manual steps. This prevents product loss.
3. **Recipe Validation:** Before setting `RecipeReady = TRUE`, the PLC should verify `ProcessCount > 0` and `BatchID` is not empty.
4. **Watchdog:** Node-RED should monitor its own S7 connection. If the S7 link drops, it should publish an alarm to RabbitMQ so FastAPI can alert the operator.

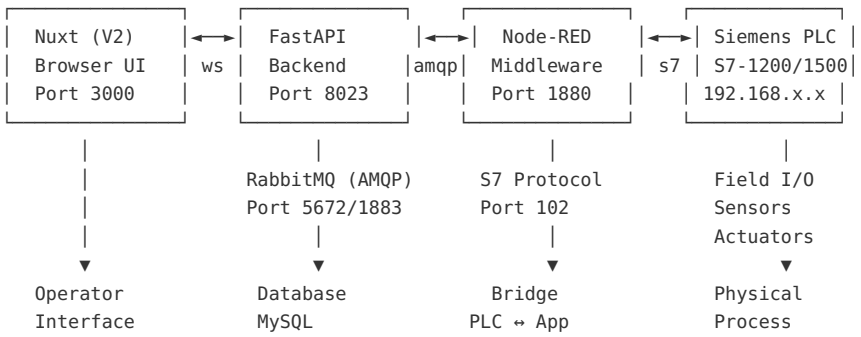
บทที่ 12: Workflow สุดท้าย

Final Workflow — PLC Mixing Control System (V2)

Version: 1.0 — May 2026
System: xMixing Control — Mitr Phol
Architecture: PLC as Executor, PC as Supervisor with Step Confirmation

---

1. System Architecture



Role of Each Component

Component	Role	Stateful?
	Executes steps, holds state, controls outputs	

PLC	(agitator, heater, valves)	Yes — **State Master**
Node-RED	Bridges PLC ↔ RabbitMQ. Polls PLC every 500ms, publishes changes	× No — Stateless bridge
FastAPI	Formats recipes, logs to DB, broadcasts WebSocket to UI	× No — Event-driven
Nuxt (V2)	Displays live state, provides confirm buttons for operator	× No — Stateless viewer
RabbitMQ	Message broker between all components	× No — Transport only

---

2. Complete Workflow — Step by Step

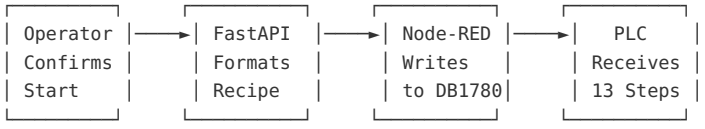
Phase A: Batch Setup (Before Production)

```
Step A1 ──► Step A2 ──► Step A3 ──► Step A4
Planning    PreBatch    Check       Navigate
            Weighing    Verify      to Mixing
```

Step	Who	Action	System
A1	Planner	Creates Production Plan with SKU, batch size, plant assignment	Nuxt → FastAPI → MySQL
A2	Warehouse	Weighs prebatch ingredients per plan, scans labels	Nuxt → FastAPI → MySQL
A3	Operator	Opens “Check for Production” page, verifies all items scanned	Nuxt reads MySQL
A4	Operator	Clicks “Start Production” → redirected to **Mixing Control V2** page	Nuxt navigation

---

Phase B: Recipe Download to PLC

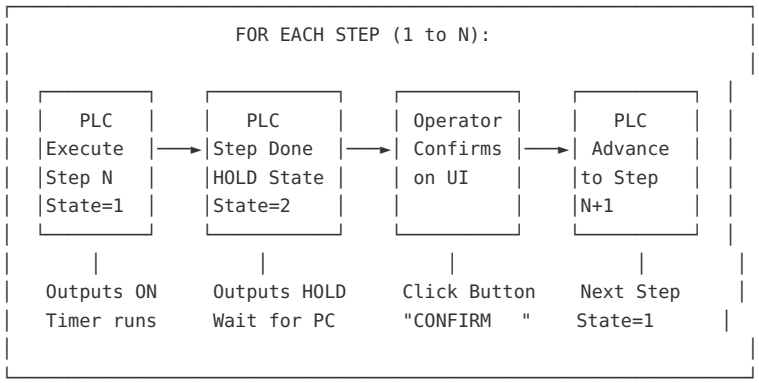


Step	Detail

B1	Operator sees “Confirm Production Start” dialog with Batch ID, SKU, Batch Size
B2	Operator clicks **CONFIRM START PRODUCTION**
B3	Frontend calls `POST /plc/send-recipe/{batch_id}`
B4	FastAPI queries `sku_steps` table → builds DB1780 payload (32 phases × 8 steps)
B5	FastAPI sends payload to Node-RED HTTP endpoint
B6	Node-RED writes recipe array to PLC DB1780 via S7 protocol
B7	Node-RED writes `RecipeReady = TRUE` to PLC
B8	FastAPI sends `StartCmd = TRUE` via Node-RED → PLC
B9	PLC validates recipe, sets `PLC_State = 1` (Executing)

Phase C: Step Execution Cycle (Repeats for Every Step)

This is the **core loop** — it repeats for all 13 steps (or however many the SKU has).



C1. PLC Executes Step (State = 1: EXECUTING)

```
PLC reads: DB1780.Processes[current_phase].Steps[current_step]

PLC sets outputs:
  → Agitator_SP    = step.AgitatorRPM
  → HighShear_SP   = step.HighShearRPM
  → Temperature_SP = step.Temperature

PLC starts timer:
  → Step_Timer counting down from step.StepTime

PLC publishes telemetry every 500ms via Node-RED → MQTT:
  {
    "PLC_State": 1,
    "Phase_ID": "p0030",
    "Step_ID": 10,
    "Step_Done": false,
    "Step_Timer": 245,          ← countdown
    "MixTank_Temp_Act": 59.8,
    "Agitator_Act": 1500
  }
```

Frontend shows: - Current step highlighted in blue with pulse animation - Live actual values (temp, RPM, weight) updating in real-time - Timer countdown displayed - Confirm button is **DISABLED** (greyed out)

C2. Step Conditions Met → PLC Holds (State = 2: WAIT_CONFIRM)

```
When timer reaches 0 (or weight/temp condition met):

PLC sets:
  → PLC_State = 2          (Wait for Confirm)
  → Step_Done = TRUE
  → Outputs HOLD at current setpoints (agitator keeps spinning)

PLC publishes:
  {
    "PLC_State": 2,
    "Phase_ID": "p0030",
    "Step_ID": 10,
    "Step_Done": true,          ← step finished!
    "Step_Timer": 0
  }
```

Frontend shows: - Step row turns **GREEN** with “Step Complete” indicator - Big **CONFIRM STEP DONE** button becomes **ENABLED** - Actual values shown next to setpoints for operator verification: - `Temperature: 60.2°C / 60.0°C` - `Agitator: 1500 RPM / 1500 RPM`

C3. Operator Confirms (Frontend → PLC)

```
Operator visually checks:
  ✓ Temperature reached setpoint?
  ✓ Weight matches requirement?
  ✓ Timer completed?
  ✓ For manual steps: ingredient scanned and added?

Operator clicks: "CONFIRM STEP DONE   "

Frontend publishes MQTT:
  Topic: mixing/plant/1/step_confirm
  {
    "Confirm_Step": true,
    "Confirm_Phase_ID": "p0030",    ← echo back which step
    "Confirm_Step_ID": 10,           ← safety: PLC validates this
    "operator": "cj",
    "timestamp": "2026-05-10T17:55:00"
  }

Node-RED receives → writes to PLC:
  DB1780.Confirm_Step = TRUE
  DB1780.Confirm_Phase_ID = "p0030"
  DB1780.Confirm_Step_ID = 10
```

Frontend also: - Logs the confirmation to FastAPI → MySQL (`mixing_batch_step_log`) - Records actual values at time of confirmation - Disables confirm button for 2 seconds (double-click guard)

C4. PLC Validates and Advances

```
PLC checks:
  IF Confirm_Step = TRUE
```

```
    AND Confirm_Phase_ID matches current phase
    AND Confirm_Step_ID matches current step
  THEN:
    → Reset: Step_Done = FALSE, Confirm_Step = FALSE
    → Advance: Current_Step += 1 (or next phase if end of phase)
    → PLC_State = 1  (back to Executing)
    → Start executing next step immediately
  ELSE:
    → Reject: Error_Code = 3 (Phase/Step mismatch)
    → Stay in State 2
  END_IF
```

Phase D: Special Step Types

D1. Manual Addition Steps (ActionCode 21010)

```
PLC executes step → PLC_State = 2 immediately (no timer)
                  ↓
Operator physically adds ingredient to tank
                  ↓
Operator scans ingredient barcode on UI (verification)
                  ↓
UI validates scan matches re_code for this step
                  ↓
Confirm button enables → Operator clicks CONFIRM
                  ↓
PLC advances to next step
```

D2. QC Recording Steps (Brix/pH Required)

```
PLC executes step → timer completes → PLC_State = 2
                  ↓
UI detects: step.BrixRecord = TRUE or step.PhRecord = TRUE
                  ↓
QC Dialog pops up: "Enter Actual Brix: ___  Enter Actual pH: ___"
                  ↓
Operator enters QC values → clicks CONFIRM
                  ↓
Frontend logs QC values to MySQL + sends confirm to PLC
                  ↓
PLC advances to next step
```

D3. Long Duration Steps (e.g., Pasteurize 300 min)

```
PLC executes step → timer counts from 300:00 to 0:00
                  ↓
During execution: UI shows live countdown + actual temp
                  ↓
Timer reaches 0 → PLC_State = 2 (Wait Confirm)
                  ↓
Operator verifies pasteurization complete → clicks CONFIRM
                  ↓
PLC advances to next step
```

Phase E: Batch Completion

```
Last step (Step 13/13) confirmed by operator
                  ↓
PLC sets: PLC_State = 4 (DONE), Batch_Complete = TRUE
                  ↓
PLC sets all outputs to safe state:
  Agitator_SP = 0, HighShear_SP = 0, Temperature_SP = 0
                  ↓
Frontend shows: "   BATCH COMPLETE!" banner
                  ↓
FastAPI updates production_batch status = "Completed" in MySQL
                  ↓
Operator can navigate back to production planning
```

---

3. Recovery Workflow (PC/App Crash)

```
Normal Operation          PC Dies at Step 7        PC Reboots
─────────────────    ──────────────────────    ─────────────────

Step 5: EXECUTING    PLC detects heartbeat    App starts
Step 6: CONFIRM      loss after 5 seconds
Step 7: EXECUTING    ──────────────────────    App reads PLC:
                     PLC continues Step 7      "Batch: P260309"
                     if AUTO (keeps timer)     "Step 7, State=2"
                     ──────────────────────
                     Step 7 timer done         UI jumps to Step 7
                     PLC_State = 2 (HOLD)      Shows CONFIRM button
                     Waits for operator...     Operator confirms
                                               Step 8 starts
```

Key Rule: The PLC **never loses its place**. It holds `Current_Phase`, `Current_Step`, and `Batch_ID` in retentive memory (DB1780). No matter how many times the PC crashes, the PLC keeps the recipe and the progress.

---

4. PLC State Diagram

```
stateDiagram-v2
    [*] --> IDLE: Power On
    IDLE --> EXECUTING: StartCmd + RecipeReady
    EXECUTING --> WAIT_CONFIRM: Step conditions met
    WAIT_CONFIRM --> EXECUTING: Operator Confirm (advance)
    EXECUTING --> DONE: Last step confirmed

    EXECUTING --> PAUSED: PauseCmd
    WAIT_CONFIRM --> PAUSED: PauseCmd
    PAUSED --> EXECUTING: StartCmd (resume)

    EXECUTING --> IDLE: AbortCmd
    WAIT_CONFIRM --> IDLE: AbortCmd
    PAUSED --> IDLE: AbortCmd

    DONE --> IDLE: New batch loaded
```

---

5. MQTT Topic Map

Topic	Direction	Purpose
`MIX-01-PUT`	PC → PLC	Heartbeat + current step info (every 1s)
`MIX-01-READ`	PLC → PC	Telemetry readback (every 500ms)
`mixing/plant/1/status`	PLC → PC	State changes, step complete events
`mixing/plant/1/cmd`	PC → PLC	START, PAUSE, ABORT commands
`mixing/plant/1/step_confirm`	PC → PLC	Step confirmation with Phase/Step echo
`mixing/plant/1/recipe`	PC → PLC	Recipe download (one-time at batch start)

---

6. Database Tables Involved

Table	Purpose
`production_plans`	Batch planning with SKU, size, plant
`production_batches`	Individual batch records
`sku_steps`	Master recipe steps per SKU
`prebatch_items`	Weighed ingredients per batch
`mixing_batch_step_log`	**Step-by-step execution log** with timestamps, actuals, operator

mixing_batch_step_log Schema:

Column	Type	Description
id	INT	Auto-increment
batch_id	VARCHAR(20)	Batch reference
phase_number	VARCHAR(10)	Phase (e.g., p0030)
sub_step	INT	Step within phase
action_code	VARCHAR(10)	Action executed
started_at	DATETIME	When PLC started this step
confirmed_at	DATETIME	When operator confirmed
confirmed_by	VARCHAR(50)	Operator username
temp_actual	FLOAT	Actual temp at confirmation
weight_actual	FLOAT	Actual weight at confirmation
agitator_actual	FLOAT	Actual RPM at confirmation
brix_actual	FLOAT	Brix reading (if QC step)
ph_actual	FLOAT	pH reading (if QC step)
duration_sec	INT	Actual step duration

status	VARCHAR(20) completed / skipped / error

7. File Map — What Goes Where

```
x2512001-mitrPhol-x31-xMixingControl-site/
│
├── x9000-Concept/
│   ├── Final_Workflow.md                    ← THIS FILE
│   ├── PLC_Recovery_Workflow.md             ← Recovery concept
│   ├── Step_Confirmation_Concept.md         ← Confirm logic detail
│   ├── Production_Deployment_Guide.md       ← PLC code + Node-RED setup
│   ├── Implementation_Plan.md               ← 3-phase dev plan
│   └── s7_1200_db_design.md                 ← DB1780 structure
│
├── x3101-app/
│   ├── x3101-0110-frontEnd/app/pages/
│   │   ├── x61-MixingControl.vue            ← Original (PC-driven)
│   │   └── x62-MixingControlV2.vue          ← NEW (PLC-master, stateless)
│   │
│   └── x3101-0210-backEnd/x0201-fastAPI/
│       ├── main.py                          ← FastAPI entry point
│       ├── routers/router_plc.py            ← PLC recipe endpoints
│       ├── plc_datablock.py                 ← DB1780 serializer
│       ├── plc_interface.py                 ← DB100/DB200 models
│       ├── mock_plc.py                      ← Quick 5-step simulator
│       └── mock_plc_full.py                 ← Full production simulator
│
├── x3109-locMqtt/                           ← RabbitMQ Docker config
└── x3112-nodeRed/                           ← Node-RED flows
```

8. Production Readiness Checklist

PLC (TIA Portal)

- ☐ Create UDT_ProcessStep, UDT_Process
- ☐ Create DB1780 (Optimized Access = FALSE)
- ☐ Create FB1780 with execute → hold → confirm → advance logic
- ☐ Add Step_Done, Confirm_Step, Confirm_Phase_ID, Confirm_Step_ID to DB1780
- ☐ Map analog I/O tags (temperature, weight, RPM sensors)
- ☐ Download and test with watch table
- ☐ Record exact byte offsets for Node-RED

Node-RED

- ☐ Install node-red-contrib-s7 + node-red-contrib-amqp
- ☐ Configure S7 connection to PLC
- ☐ Create polling flow (read telemetry every 500ms)
- ☐ Create change-detection function
- ☐ Create RabbitMQ publish node
- ☐ Create recipe write flow (HTTP → S7)

- [ ] Create confirm relay flow (MQTT → S7)
- [ ] Create heartbeat relay flow

FastAPI Backend

- [ ] Update /plc/send-recipe to POST to Node-RED
- [ ] Add RabbitMQ consumer for plc.mixing.state
- [ ] Add WebSocket broadcast to Nuxt
- [ ] Add mixing_batch_step_log table + insert on step confirm
- [ ] Add heartbeat publisher (toggle every 1s)

Nuxt Frontend (x62-MixingControlV2.vue)

- [x] Stateless UI — reacts to PLC telemetry only
- [x] Recovery on mount (reads PLC state on page load)
- [x] Recipe download on Confirm Start
- [ ] Add “CONFIRM STEP DONE” button (enabled when PLC_State = 2)
- [ ] Add QC dialog for Brix/pH steps
- [ ] Add step confirmation MQTT publish with Phase/Step echo
- [ ] Add double-click guard (2s disable after confirm)
- [ ] Log confirmation to backend API

Testing

- [ ] Run mock_plc_full.py — verify all 13 steps appear in UI
- [ ] Test step confirmation flow end-to-end
- [ ] Pull-the-plug test: kill FastAPI at Step 7, restart, verify recovery
- [ ] Manual step test: verify PLC holds on ActionCode 21010
- [ ] QC step test: verify Brix/pH dialog appears and records values
- [ ] Double-confirm test: verify rapid clicks don’t skip steps